



LINCOLN INSTITUTE
OF LAND POLICY

From Vision to Value: A Case Study of How Seven Danish Cities Conduct Area Development to Propel Urban Revival

Working Paper WP21LN1

Luise Noring
City Facilitators

June 2021

The findings and conclusions of this Working Paper reflect the views of the author(s) and have not been subject to a detailed review by the staff of the Lincoln Institute of Land Policy. Contact the Lincoln Institute with questions or requests for permission to reprint this paper. help@lincolninst.edu

© 2021 Lincoln Institute of Land Policy

Abstract

This working paper presents how large-scale urban redevelopment across seven Danish cities and towns is delivered and financed using land value capture. In all the seven cases, partially or fully municipal development corporations play a critical role in leveraging publicly accumulated and owned land and buildings to spur local economic growth.

The present working paper considers development corporations or equivalent public, private and civic hybrid entities that were selected based on their geographical distribution, including in cities and densely urbanized areas (Housing and Project Development in Aarhus, Stigsborg Havnefront in Aalborg), small towns (Nærheden in Høje-Taastrup, Køge Kyst in Køge, Holstebro Udvikling in Holstebro, Kanalbyen in Fredericia), and a rural area (Ringkøbing-Skjern).

The cities presented are driving transformational visions and associated initiatives and investments that remake their physical, economic and social landscape. These visions are driven by forward thinking city governments that use their partial or full ownership of the development corporations as institutional vehicles. The development corporations are able to catalyze long-term financing that aligns with the breadth and depth of the city vision. In some of the cases, the long-term horizon is backed by philanthropic patient capital that enables the cities to leverage land value capture for continuous reinvestment and revitalization. Such a tried and tested method evidenced by the case studies presented here renders an opportunity for cities across the world to embark upon similar trajectories.

About the Author

Dr. Luise Noring is an Assistant Professor at Copenhagen Business School. She has a Ph.D. in supply chain partnerships and is a former expert member of the European Commission on renaturing cities and of the OECD on land value capture.

Dr. Noring is a recognized expert on urban governance and finance, with a particular focus on how cities deliver and finance urban regeneration, infrastructure projects, affordable housing, and sustainable economic growth. Noring specializes in codifying innovative institutional models and finance mechanisms that drive environmentally sustainable and socially inclusive cities. Her body of research has been published in respected peer-reviewed academic journals, such as *Urban Studies*, *Urban Affairs Review*, *Cities*, and *Urban Research & Practice*.

She has authored or co-authored case studies on a broad array of pathbreaking solutions, including the Copenhagen City and Port Development Corporation, the Danish Cooperative Housing Model, the Land Value Capture experiences across seven Danish cities and towns, and the Cincinnati Centre City Development Corporation. This body of applied research has been published by City Lab, the Brookings Institution, the Lincoln Institute, ICLEI, and the Nowak Metro Finance Lab at Drexel. Noring's research reveals how cities can conduct large-scale transformative investments without relying entirely on public funds. Rather cities are encouraged to deploy underutilized public assets for land value capture, self-organize and self-finance affordable and social housing in cooperatives, and leverage municipal financing through mutually reinforcing the collective power of cities.

Since 2016, Dr. Noring and her company, City Facilitators, has advised public, private and civic leaders on delivering City Solutions for sustainable and inclusive economic development. With the City Solutions, Noring and City Facilitators bring depth and specificity to the analysis of urban governance and finance, distilling enabling features and financial mechanisms, to enable large-scale adoption and adaptation. City Facilitators has delivered vanguard City Solutions to over 25 public, private and civic clients across more than 15 countries.

Acknowledgments

This endeavor would not have been possible without the special help and guidance of key stakeholders across the seven Danish cities and towns, Realdania, and Copenhagen City & Port Development:

Name	Title	Company and municipality
Claus Ravn	Lead Consultant	Realdania By & Byg
Ole Møller	CEO/ Project Director	Nærheden P/S, Høje-Taastrup
Bente Lykke Sørensen	City Strategic Director	Aarhus Municipality
Michael Tolstrup	Head of Department	Aarhus Municipality
Mia Hørup	Special Consultant	Aarhus Municipality
Claus Omann Jensen	CEO / Director	Færchfonden, Holsterbro
Hanne Torp Brodersen	Project Director	Færchfonden, Holsterbro
Peter Frost	Municipal Director	Køge Municipality
Tove Skrumsager Frederiksen	Project Director of Køge Kyst P/S	Køge Municipality
Hans Østergaard	Mayor	Ringkøbing-Skjern Municipality
Anton Hessellund	Director	Stigsborg Havnefront P/S, Aalborg Municipality
Tim E. Halvorsen	CEO / Director	Naturbydelen in Ringkøbing + Kanalbyen in Fredericia
Peder Baltzer Nielsen	City Architect	Aalborg Municipality

Table of Contents

Introduction	1
Main Findings	3
Matrix Overview.....	5
Map	8
Danish Development Corporations	9
<i>Køge</i>	9
Introduction.....	9
Sequencing.....	9
Strong Vision	12
Ownership	15
Organizational structure	15
Investment.....	16
Access to Finance.....	17
Conclusion	18
<i>Høje-Taastrup</i>	18
Introduction.....	18
Sequencing.....	19
Strong Vision	20
Organizational Structure	24
Ownership	25
Investment.....	25
Access to Capital.....	25
Conclusion	26
<i>Holstebro</i>	27
Introduction.....	27
Sequencing.....	29
Strong Vision	31
Organizational Structure	32
Ownership	32
Investment.....	33
Access to Finance.....	33
Conclusion	33
<i>Fredericia</i>	34
Introduction.....	34
Sequencing.....	34
Strong Vision	36
Organizational Structure	39
Ownership	39
Investment.....	39
Access to Finance.....	40
Conclusion	40
<i>Ringkøbing-Skjern</i>	41
Introduction.....	41

Sequencing.....	41
Strong Vision	42
Organizational Structure	45
Ownership	46
Investment.....	46
Access to Finance.....	46
Conclusion	46
<i>Aalborg</i>	47
Introduction.....	47
Sequencing.....	49
Strong Vision	50
Organizational Structure	51
Ownership	54
Investment.....	55
Access to Capital.....	55
Conclusion	56
<i>Aarhus</i>	56
Introduction.....	56
Sequencing.....	57
Strong Vision	60
Organizational Structure	61
Ownership	61
Investment.....	61
Access to Finance.....	64
Conclusion	65
Conclusion	66
<i>A Strong Vision</i>	66
<i>Land Value Capture</i>	66
<i>Replication in Other Cities</i>	67
List of Interviewees.....	68

From Vision to Value: A Case Study of How Seven Danish Cities Conduct Area Development to Propel Urban Revival

Introduction

As the global recession, climate change, the pandemic to name but a few globally transformative events blow through the world deep rooted inadequacies and malfunctions of our societies are exacerbated. In fact, such global events reveal good as well as bad societies. These events are compelling cities to rethink how we design, finance and deliver urban redevelopment. The pre-pandemic model was neither inclusive nor sustainable. In Denmark, an alternative model has emerged that uses the disposition of public assets to drive the creation of public wealth.

The pre-pandemic development model had three main deficiencies. First, growth was unusually concentrated in a small number of large superstar cities that dominated national economies (e.g., Copenhagen, London, San Francisco) and garnered the lion's share of private investment. This spatial imbalance placed smaller, weaker market cities at a decided disadvantage, and they struggled to attract and retain companies and residents.

Second, private capital, oriented toward these over-heated cities, demanded excessive returns, making redevelopment in weaker market (often former industrial) cities difficult to finance without large amounts of public subsidy. Private capital, even when philanthropic in nature, had little incentive to innovate and evolve to meet the needs of communities that had prospered in prior decades.

Finally, the benefits of growth redounded almost exclusively to private investors and interests, even though public assets and investments often provided the foundation for project finance. The appreciation of value that generally followed project completion was rarely captured for public use and reinvestment.

This working paper grants us the opportunity to understand how urban renewal and redevelopment have been conducted across seven Danish cities and towns. We learn how new institutional vehicles and novel collaborative constellations can transform cities and towns of different geographies, sizes and challenges. Collectively, these cities and towns have managed to turn the tide and reverse economic decline and depopulation.

The development corporations represented in this working paper are either wholly or partly municipal-owned. A common feature of all the development corporations is that they have a strong vision that permeates the developments and enables the organizations to compete in niche markets. The wholly or partly publicly owned and privately driven development corporations driving urban redevelopment across these seven Danish cities and towns offer an alternative path that allows city governments to steer developments in close alignment with the broader political and economic vision of the city and the concrete areas under development, while harvesting benefits provided by the market. Typically, the ownership of joint ventures is split 49 percent to the municipality for contributing land assets and 51 percent to private investors for contributing capital investments.

There are strict rules (read *kommunal fuldmagten*) preventing Danish municipalities from engaging in commercial developments. These are linked to Danish laws that hinder municipalities from making speculative investments and distorting the market through their advantage of both owning the land assets and controlling the planning, zoning, and permits. Municipalities are only allowed to develop buildings that are destined for municipal usage.

The way the municipalities proceed is to attain a permission for the establishment of part-publicly owned and privately driven development corporations. The municipalities seek legal advice to form the corporations jointly owned with private and/civic partners. This legal documentation is submitted to the National Appeals Board (read *Tilsynsrådet*), which grants permission to establish the part-publicly owned corporations guided by private market laws. This means that albeit the corporations (read *P/S*) being part-publicly owned by the municipality, they are not obliged to disclose information publicly nor conduct tenders and public hearings, which are obligations normally ascribed to municipalities. This does not mean that the part-publicly owned development corporations always refrain from conducting tenders and public hearings, but it does mean that they can do so at their own discretion. Generally speaking, in all the part-publicly owned development corporations presented in this working paper, the corporations conduct public tenders for private developments.

The sales of the development projects generate revenue that is channeled back into infrastructure and other investments, such as soil cleansing, climate protection and recreational areas. In some cases, the revenue generated is substantial is used to finance broader infrastructure investments beyond the areas under development. This model enables municipalities to conduct large-scale catalytic urban redevelopment without spending taxpayers' scarce money that can be dedicated to other productive purposes, including purchasing land for urban redevelopment and financing infrastructure investments. This tried-and-tested model of wholly or partly publicly owned and privately-run development corporations has the potential to propel towns and cities across the globe onto a positive trajectory of transformative urban regeneration and economic revitalization.

Realdania By & Byg

In this working paper, Realdania plays a central role. Realdania is “a philanthropic association with some 165,000 members” that undertakes “philanthropy on the basis of returns on its investment assets,” (1) with a focus on improving the built environment. It is Realdania’s for-profit arm Realdania By & Byg that is the partner in four of the cases considered in this working paper. Across all the four cases, Realdania By & Byg operates for profit, but in markets that have no proved business case. Realdania By & Byg operates in weak markets with the purpose of demonstrating that even in such markets, it is possible to generate profits, if developments are anchored in a strong vision and investments are conducted over a long-term investment horizon.

Realdania By & Byg’s stated mission is to leverage the parent association’s assets in order to:

¹ Realdania (n.d.b). We are a philanthropic association. *Realdania.dk*.
<https://www.realdania.org/whoweare/wearephilanthropicassociation>.

“carry out new builds and pursue construction experiments at full scale and to develop and realize the visions for urban life of the future via development corporations with Danish municipalities and other investors. [...] When Realdania By & Byg invests in properties and areas, it is in order to develop or ensure qualities that would otherwise be lost, and we typically only get involved in projects that others cannot support. It is also a precondition that the use is current and that operations are financially sustainable”.²

Main Findings

The below list of findings are key enabling features that US cities and towns can adapt to replicate the model presented in this working paper.

A Strong Vision

All seven development efforts studied in this working paper possess strong and unique visions. These visions were coined by the development corporations themselves, sometimes in collaboration with the local city leadership, other times in collaboration with the investors or local citizens. These visions serve multiple purposes:

1. Strong visions can shield developments from the vagaries of municipal politics. New political leaders may set new agendas with different political priorities. This can impact development projects in several ways, including by hampering existing relationships with investors that have bought into the vision, which can prevent developments from moving forward as intended. Bearing in mind that new leadership can also potentially have a positive impact on developments, it is important that investors know that the original promises, plans, and prospects will be followed regardless of who is in charge. This vision helps keep developments on a steady course.
2. A powerful vision can allow a development to compete in niche markets as opposed to competing on the basis of price alone. This makes it possible to differentiate developments in highly competitive markets. Because investors have an abundance of potential projects they can pursue, development corporations can position themselves in less crowded markets, where price is not the only determinant of land value.
3. Vision-driven markets encourage experimentation. Having a strong vision encourages development corporations, investors, and developers to push market boundaries through experimentation. Reinforcing the vision through deployment of cutting-edge solutions and new product and service innovations strengthens the investment proposition. Developments become known for their fluid integration of urban functions, their art and architecture, their opportunities for cross-generational co-living, or their novel forms of work-life balance that combine low-maintenance housing with an abundance of recreational offerings.

² Realdania (n.d.b). We are a philanthropic association. *Realdania.dk*.
<https://www.realdania.org/whoweare/weareaphilanthropicassociation>.

Bundling of Publicly Owned Assets

By bundling assets, development corporations are able to reach critical mass, which serves multiple purposes:

1. Bundling assets enhances transformative impact. When a development area is large relative to its town or city, the development can have major implications for the area. A single large development project can propel an entire town's revitalization. Towns can take on entirely new identities, for example, attracting artists, drawing eco-tourists, hosting sports activities, and showcasing restaurants and an abundance of street-level activities.
2. By bundling assets, development corporations can attract large institutional investors. Large investors often target large projects with hundreds of residential units because this allows them to achieve economies of scale, including benefits from lower unit construction costs, and to make a greater impact on the area through a large, iconic building.
3. Bundling assets boosts economies of scale in infrastructure. A large-scale project makes it easier to justify large investments in expensive infrastructure, such as new power, water, IT, or sewage systems; new roads; new schools; and new sports and leisure facilities. Municipal governments must provide this reasoning in their defense of spending taxpayers' money on large public investments. There must be demonstratable evidence of real need for such investments, which can, when managed correctly, have a substantial impact on the value of entire development areas.

Attraction of Institutional Investors

With a strong vision and a large-scale development, development corporations are able to attract institutional investors, which provides numerous benefits:

1. The presence of institutional investors offers financial stability and security for large-scale developments. Pension funds are important institutional investors in Denmark, and in several other countries, like the United States, with well consolidated finances that can withstand nearly all market fluctuations.
2. Institutional investors enable the realization of long-term visions. Pension funds invest with a 30- to 40-year horizon, which means that they are willing to invest in quality developments that can withstand the passage of time and that they appreciate the value of a strong vision that translates into a unique brand for the area, which can in turn attract particular segments of the population (young families, cross-generational families, nature lovers, etc.).
3. Institutional investors can help leverage other investors. Once pension funds are investing in an area, other private investors are encouraged to also invest in the area. Danish pension funds are known for their conservative investment strategies, high-quality building standards, and long-term engagement, features that serve as a guarantee for other private investors and developers that follow in their wake.

Matrix Overview

The below matrix overview of Danish urban development corporations provides a visual comparison of different kinds of urban development organizational models.

	Challenges	Opportunities	Key Enabling Features
Køge (Coastal town with a population of 38,000 located 39 km (45 minutes by train) south of Copenhagen)	Railway tracks divide the new development areas from the old town center, making it difficult to create a coherent and unified town.	Three new crossings are crucial for Køge Kyst's development. One crossing is a 15-meter-wide pedestrian overpass funded by Realdania Filantropi and Realdania By & Byg. Another crossing is a pedestrian bridge that doubles as a park, which opened in 2017.	The Køge Kyst development is a stellar example of long-term and continual dedication to a vision for an attractive suburban town. Considerable investments were made in connectivity between the new development area and the old town as well as in creating attractive new recreational areas along the coast that invite residents to make use of the new area.
Høje-Taastrup (The suburban municipality of Copenhagen has an aging population of around 51,000)	The neighborhood adjacent to the Nærheden redevelopment area hosts single-family homes from the 1970s. The municipality uses the publicly owned area of Nærheden to propel an influx of resourceful young families and elderly people.	The development corporation, which is coowned by the municipality and Realdania By & Byg, decided to create an area offering new forms of intergenerational living. This translated into diverse living and housing arrangements, including small homes with high levels of external services, and homes for the elderly with a large indoor communal garden.	The development is at the forefront of new solutions for life, work, and family, especially those involving intergenerational living.
Holstebro (A town of 36,600 people in Northwest Jutland peripherally)	The town was struggling with depopulation and economic decline, coupled with a lack of	The Færch Foundation and the town worked together to first envision a new future and then to move	This partnership demonstrates how philanthropic foundations around the world can contribute to

located relative to Denmark's economic, political and population centers)	municipal institutional and financial capacity.	toward its realization. Realdania Filantropi had already invested in a new state-of-the-art Talent Academy to retain and attract young people.	meaningful and high-impact urban development.
Fredericia (The town of around 41,000 in East Jutland is a former industrial center of ship building with substantial historic heritage)	The town faces fierce competition from other towns in Denmark's Triangle Region, which have likewise experienced deindustrialization resulting in vacant waterfront land and buildings. The town must furthermore consider noise and soil pollution.	The former industrial harbor transformed an area with a strong historic narrative into open water canals and integrated works of public art that communicate the town's story.	The town embraces the waterfront and harbor, rather than turning its back to it. It implements a strong vision that permeates every aspect of the development, including integrated art installations in buildings and public areas.
Ringkøbing-Skjern (With a total population of around 57,000, it is a small rural town in a geographically extensive municipality donning an abundance of nature)	The town had experienced a decade of stagnancy, with worrisome demographic trends and slow economic growth.	Rather than selling its assets to the highest bidder, the town and its development corporation embarked upon an experimental effort to create a mono-functional residential area with a strong and unique vision, making the most of the area's natural beauty, with its meadows and seascapes.	Most development projects embrace mixed-used development, but markets that have struggled to attract newcomers and investment must consider fresh propositions. In such markets, it may actually be a better strategy to offer a mono-functional development with a strong and unique message.
Aalborg (The third largest city in Denmark with a population of around 211,000 has a strong industrial legacy)	The three joint venture partners have different objectives for the partnership and different approaches to urban development.	The three partners made minor but high-impact adjustments to the project's governance model, which included bringing key staff closer to the local municipal administration and establishing a cross-	The ownership model involves three capital partners that collaborate with one another while fulfilling their own distinct needs. This unusual ownership model showcases the unique strengths and resources of different kinds of project

		organizational taskforce for troubleshooting. This helped realign the project and renewed support for its long-term vision.	partners: 1) the municipality possesses the land and holds zoning competences and planning skills; 2) A. Enggaard is a private developer with vast experience in large-scale development projects; and 3) PFA Pension has funds for long-term investment.
Aarhus (Greater Aarhus currently has a population of around 350,000 making Aarhus the second largest city in Denmark. The city holds a large still functional harbor)	The new developments coincided with the global recession. Thus, most of the existing developers had gone bankrupt by the time construction was scheduled to commence.	The project has benefited from close collaboration between the municipality (which sets the vision) and private investors (which invest in and construct distinct projects). The challenge is that the municipality must handle both the role of political visionary and the role of project developer that generates economic value.	The organizational structure is an internal municipal government administrative entity (35 employees) that is dedicated entirely to the redevelopment of the five selected areas. The entity has perfected the process of mediating between municipal and investor/developer interests and actions, helping to translate vision into economic value.

Table 1: Overview of development corporations presented in this working paper. Source: Noring, 2020

Map

The below map shows the different urban transformative developments investigated in this working paper:

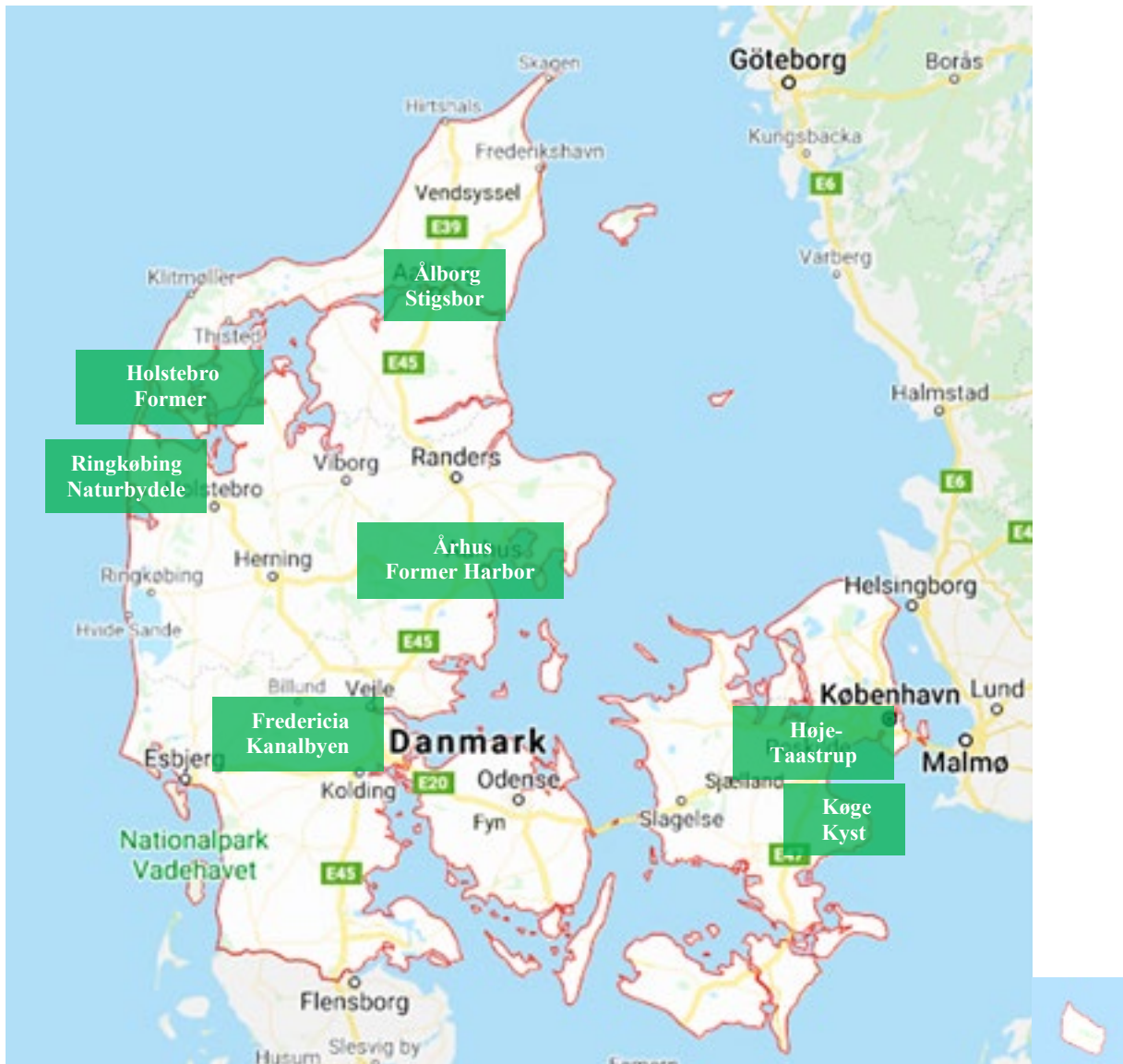


Image 1: Map of Denmark. Source: Google Maps, 2020.

Danish Development Corporations

Køge

Introduction

The first case study, Køge Kyst, is among the largest redevelopment projects in Denmark. Køge Kyst consists of 5.6 hectares around the train station, 15.2 hectares in the South Harbor, and 3.1 hectares in Collstrop, totaling 24 hectares and 300,000 floor m² of new residential, retail, and office construction.



Image 2: Køge Kyst development areas. Source: Jan Kofod Winther, 2020.

The development zone includes land that has been reclaimed in the harbor area. A fully integrated climate protection plan, launched in 2015, includes widening the natural beach, restoring the natural wetland area, constructing a promenade that doubles as a flood barrier, raising the ground level of the buildings, and creating green wedges within the built-up area to collect flood water. It is a fully integrated climate protection plan that largely relies on natural elements. This plan has been jointly led by the municipality and the national Nature Preservation Authority.

Sequencing

In recognition of Denmark's changing economic landscape and the country's shift from a manufacturing to a service economy, in 1995, local politicians, municipal employees, business leaders, and residents in Køge decided to rethink the town's objectives. During this period, early

ideas centered on how to connect the industrial harbor and waterfront and the old town center, which were separated by railway tracks. However, several years passed before any action was taken to revamp the town's economy. Industry was still operating in the harbor, but once leases expired, the municipal-owned Port Authority declined to renew them. This meant that the land was gradually being freed up for redevelopment.

Between 1995 and 2007, the municipality worked on developing the industrial harbor, but the task was perceived as too complex for the municipality to manage without a partner due to soil contamination, climate protection needs, and the necessity of making a return on investment. The municipality thus chose to partner with Realdania By & Byg.

In 2004, construction commenced on a new industrial harbor to the north of the old harbor. The old harbor was too narrow for large ships to enter. The municipal-owned Port Authority established the Køge Jorddepot (read Køge Land Holdings) company, which was responsible for land reclamation activities. The blue area in the image below represents the new development zone in South Harbor that formerly housed the industrial harbor. The orange area represents the new industrial harbor built on reclaimed land.



Image 3: Køge: The orange area represents the new industrial harbor that is freeing up land for development in the former industrial harbor, which is represented by the blue area. Source: Jan Kofod Winther, 2020.

The railway tracks presented a significant infrastructural challenge. In addition, the former industrial land raised concerns about pollution. In 2000, the municipal government embarked on an exercise to determine the best institutional form to lead a development project of this complexity.

First though, the municipality had to apply for permission from the National Supervisory Authority to establish a development corporation that would operate at an arm’s length from the local government and be jointly owned by the municipality and Realdania. This arm’s length relationship is exemplified by the development corporation not taking dictate from the political leadership of the municipality, and therefore, also transcends shifts in political leadership and priorities, which enables the development corporation to stay on track with its’ long-term vision of development. The municipality received permission to establish a development corporation in 2008.

The eventual Køge Kyst P/S development corporation was innovative in the Danish context, as it was the first development corporation to be jointed owned by a municipality and Realdania By & Byg. “Realdania wanted to set a new example and standard for how to do urban redevelopment in small cities across Denmark,” Municipal Director Peter Frost said (interview, 30 June 2020).

Because smaller towns sometimes struggle to attract outside investments, they often end up compromising on what is best for the town in the long term. Thus, towns may feel inclined to construct expensive private housing in former port areas, as this is what sells best with private investors and developers. However, this runs the risk of building areas devoid of street life, atmosphere, and buzz as expensive private housing does not in itself create vibrant street life. The present working paper demonstrates the importance of taking the necessary time to create a coherent vision that serves as a key competitive parameter for the long term. It also demonstrates the importance of embedding this vision in the detailed local planning of the new development area.

In 2010, Køge Kyst P/S ran a parallel competition. Proposals by Vandkunsten and SLA (e.g., both private consulting firms) were selected, and Køge Kyst P/S requested that these two firms join forces and collaborate on a proposal that was ultimately purchased and integrated into the Køge Kyst master plan in 2011. Project developments began in 2012, when the first private sale was realized near the town center proper. Actual construction did not begin until 2014.

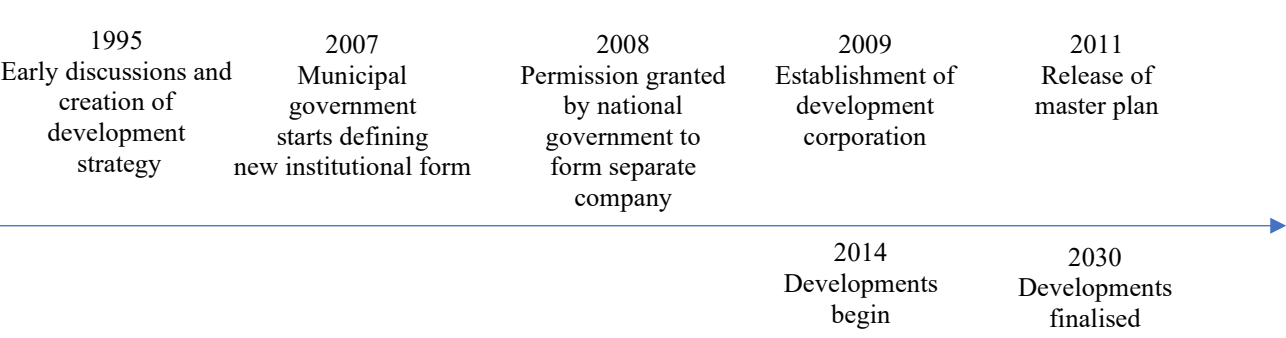


Image 4: Timeline of Køge Kyst development. Source: Noring, 2021.

The municipal government’s positive experiences with the Køge Kyst development project prompted it to release a master plan for a new development district, Køge North, in 2019. In essence, having acquired insights into how to conduct large-scale urban redevelopment, the municipality decided to head the Køge North development project itself rather than entering into a

partnership with a private investor or developer. “The municipality has been saving money for years to be able to buy up the land in Køge North. This development has been in the making for decades,” said Frost, continuing, “We said no to large investors because we wanted to pursue our vision and plan for Køge North” (interview, 30 June 2020).

In line with Køge’s expansion, the Danish public rail service discovered the potential of making Køge North the new multimodal passenger transit hub for southern Greater Copenhagen. Passengers from Køge North will be able to reach Copenhagen in 20 minutes, and passengers from Køge City will be able to do so in 22 minutes. This has in turn made the town even more attractive to new residents and investors.

The master plan took four years to complete, and the municipal government now has two accepted tenders for development in Køge North. Between the two tenders, Køge North will soon provide space for offices, retail establishments, and a hotel. The Køge North land is adjacent to the national and regional train tracks, and its value will increase considerably as a result of its vicinity to a new train station.

Strong Vision

Culture is a cornerstone of the Køge Kyst vision and is a driver for all the Køge developments. “From the start, we dedicated a large amount of money to culture. Before new development begins, we conduct temporary activities, such as activity days, fishery days, street food, bee homes, and so on, so people can claim ownership of the areas. The town needs to be buzzing with life before, during, and after the development work,” Peter Frost said (interview, 30 June 2020).

Another element of the Køge Kyst vision is the continued expansion of the town at a human scale. Frost says, “We create new builds that are in harmony and at scale with the old town. We respect the town’s history and heritage. Historically, Køge is an important town that has had big achievements. We’re holding onto our DNA. We’re holding onto our history while looking ahead to the future” (interview, 30 June 2020).

The developmental vision centers around three main key messages:

1. The old town and the industrial harbor should be connected so as to integrate the old and the new, driving Køge’s role as a thriving part of Greater Copenhagen.
2. Connectivity within Køge’s town center should be reinforced through the creation of three new crossings bridging the railway tracks that better integrate the old and the new towns.
3. New recreational and cultural activities should be developed, including the establishment of a new beach and new youth community hall.

The new crossings are critical for Køge Kyst’s development. One of the crossings is a 15-meter wide pedestrian overpass funded by Realdania Filantropi and Køge Kyst. This is planned to open in 2024.



Image 5: The 15-meter-wide pedestrian overpass connecting the city center and the newly developed area of Køge Kyst. Source: Schønher, 2020.

Another crossing is a pedestrian bridge that doubles as a park (see photo below) and opened in 2017. This crossing goes directly into the Colstrup area, making that area more accessible and thereby more attractive. It also helps improve access to the train station.



Image 6: Source: Luise Noring, 30 June 2020.

A car tunnel leading into Køge Kyst opened in 2016. The tunnel helps divert traffic away from the town center and the area connecting the center with the new residential and commercial harbor zones. There used to be a heavy traffic road alongside the railroad tracks, which combined to highlight the divide between the old town and the new harbor area. This road has been narrowed, creating space into which the old town center's retail district can expand.

In the below figure, the wide blue area represents the original road infrastructure. The orange area represents the new condensed road running directly alongside the rail tracks.



Image 7: The blue area depicts the original road infrastructure. The orange area shows the path of the new road infrastructure. Source: Rune Pedersen, 2020.

All these projects have been financed by Køge Kyst P/S using revenues from land sales. The pedestrian crossings help create a flow and circulation of people in the dense retail area. “We could easily have put up a large shopping center, but that would have killed street life,” Frost says (interview, 30 June 2020). The Køge Kyst P/S development corporation thus determines what the ground floors of new construction must offer. These stipulations are then written into sales agreements.

Even though the town’s municipal administration is not overseeing the Køge Kyst developments, Frost explains that developers and investors are subject to high levels of scrutiny: “The development corporation gets into every little small detail before it sells land” (interview, 30 June 2020). This is in order to fulfil the quality objectives set in the quality plans for each Køge Kyst development project. The quality plans cover environmental sustainability, community, construction quality, and architecture. Køge Kyst relies upon these quality plans to manage quality in its developments. For example, as illustrated in image 8, the corporation has decided that all surface rainwater in Køge Kyst must be collected and channeled out into the green natural environment surrounding the residential housing. The development corporation has also opted for the expensive choice of high-quality materials.



Image 8: Source: Luise Noring, 30 June 2020.

These residential developments meet the DGNB (*deutsche gesellschaft für nachhaltiges bauen*; read German Society for Sustainable Building) Gold standard³, which is a European construction standard that was originally developed in Germany. It is characterized by including the entirety of the construction cycle from material sourcing to final building construction. The residential developments also possess considerable diversity of residents: out of 80 residential units, 30 are dedicated to elderly residents, and the remaining 50 units are rental apartments of varying costs and sizes. According to Danish planning regulation, all new built development must pertain min. 25% affordable housing, of which 25% must be dedicated social housing.⁴

Ownership

The Køge Kyst P/S development corporation is half owned by Køge Municipality and half owned by Realdania By & Byg.

Organizational structure

To ensure that the corporation operates at an arm's length from the municipal government, the owners established a professional board that operates independently of the political leadership of the municipality. Additionally, this is reinforced with the stipulation that the chair will have no current or former relationship with the municipal government and the right to select the six remaining board members will be split evenly between the municipal government and Realdania By & Byg.

³ <https://www.dgnb-system.de/en/system/index.php>.

⁴ <https://iclei-europe.org/publications-tools/?c=search&uid=gDCrxJk2>.

Investment

The Køge Kyst P/S development corporation is tasked with preparing former industrial land for residential and commercial development and for financing all public infrastructure in the development areas. Due to the considerable expenses incurred during the first seven years of development, both owners have been obliged to inject additional funds on two occasions, once during the initial start of the developments and another subsequently to finance infrastructure investments. The alternative was to wait to make the infrastructure investments until land sale revenues had picked up. However, there was broad consensus that early investments in infrastructure would raise land values, and there was a reluctance to slow down developments once the corporation, vision, master plan, and so on were in place.⁵

DKK 420 million (approx. \$20 USD) have been invested as startup capital in Køge Kyst. This investment stems from Køge Municipality and Realdania By & Byg. The startup capital enabled Køge Kyst to clean the soil as well as undertake climate protection and infrastructure investments before selling off the land. These initial, crucial investments raised land values significantly of both the matured land and the land adjacent to the matured land. “In some instances, we did not make the investments before selling the land, but we built it into the sales agreement that the investments would be made,” says Project Director of Køge Kyst, Tove Skrumsager Frederiksen (interview, 10 December 2020).

Realdania By & Byg’s involvement in the development corporation provides reassurance and the baseline investment that has encouraged Danish pension funds to enter new markets. “At the start of our development, we weren’t able to attract the big institutional investors,” says Peter Frost. “Pension Danmark (a membership owned cooperative pension corporation of 770,000 members)⁶ will make the area attractive because it’s investing for the long haul. It wants a community hall, kindergarten, and so on” (interview, 30 June 2020). Pension Denmark has already built 80 rental apartments in Køge Kyst and plans to build 500 more between 2020 and 2025. The plan for the 500 new residential units will, for example, provide DGNB Gold certified structures, publicly accessible ground-level areas, and housing for the non-traditional family structures, such as parents living apart but sharing custody of children.

Køge Kyst hosts the largest cooperative housing project in Denmark, which is building its own private ownership apartments, Fællesbyg Køge Kyst. This effort is part of a collaboration with Køge Kyst, Fonden for Billige Boliger, and Realdania.

⁵ Jørgen Nue Møller. Køge Kyst: En Historie om the Bys Forandring. KAB Fonden. 2018.

⁶ <https://www.pensiondanmark.com/en/>.



Image 9: The private housing cooperative "Fællesbyg Køge Kyst." Source: Realdania By & Byg, 2020.



Image 10: Photo of the artificial beach that also serves as floor protection. Source: Luise Noring, 30 June 2020.

A promenade runs alongside the residential housing and to the new beach, which opened in 2015. Both the concrete promenade and the large beach provide the area more resilient to climate factors including provides a large floor plain protecting the buildings from stormwater and raising sea levels.

Access to Finance

Køge Kyst uses the proceeds of the land sales to pay for public infrastructure in the development area. This infrastructure includes two of the three crossings bridging the old town and the new development areas. As a result, the development corporation is under pressure to hold large cash reserves. The continued sale of land to private developers and investors is a prerequisite for the development's success, but the development corporation operates on the assumption that it will first achieve a positive bottom line in 2024.

Following construction of infrastructure such as roads, sidewalks, squares, and other public areas, the municipality will take over responsibility for maintenance. Thus, while the development

corporation Køge Kyst finances all the infrastructure, its maintenance is financed and delivered by the municipality.

Although the development corporation is not obliged to run public tenders, it tenders for all for-profit land sales in order to optimize price levels. Frost says, “We started sales at the beginning of the Global Recession. It was impossible to sell. Today, however, all residential projects are sold right away” (interview, 30 June 2020). Køge Kyst currently boasts the highest m² prices in Greater Copenhagen’s southern suburbs, including Høje-Taastrup.

Conclusion

In 2009, Realdania By & Byg entered into a partnership with the Køge Municipality to transform the deindustrialized harbor. The harbor was both separated from the town proper and had been exposed to significant environmental damage due to extensive industrial usage and heavy rail infrastructure. The entire redevelopment project is expected to take 20 years. The South Harbor still hosts some industry, which is gradually moving out to a modern port outside the urban center. The South Harbor is increasingly dedicated to housing and offices, while the area in proximity to the station is dedicated to individual small shops along the new and old streets, and the Collstrop Parcel is designated for commercial purposes, including small-scale urban manufacturing, offices, and workshops.

The initial investments by the municipality and Realdania By & Byg were used to leverage private investment. The challenge was to remodel the medieval trading town of Køge in such a way that its residents and visitors would embrace the new development area. For this purpose, three new crossings over rail lines are critical for Køge Kyst’s development, of which one of the crossings is a 15-meter-wide pedestrian overpass funded by Realdania Filantropi and Køge Kyst P/S. In much the same way, the municipality is making an additional investment in the car tunnel, which is financed by both the municipality and Køge Kyst.

Høje-Taastrup

Introduction

Høje-Taastrup is a suburban municipality in the western part of Greater Copenhagen, an area with a high concentration of social housing and ethnic minorities. The population is expanding, particularly among the elderly (80+ years). In January 2020, the population of Høje-Taastrup was 50,759, a figure that is estimated to grow to 62,170 by January 2032.

The Nærheden district (65 hectares) is undergoing redevelopment and focuses on providing greater community engagement and convenient daily life opportunities with shopping, schools, green recreational areas, and walking and biking paths that connect with the adjacent and socially diverse Hedehusene neighborhood. Hedehusene is a slightly dated development that has not experienced population growth for the past decade. The Nærheden district is located adjacent to the Hedehusene regional train station, 27 km from Copenhagen Central Station.

In 2019–2020, a new bridge was constructed across the rail tracks to connect Nærheden with Hedehusene and to serve as an architectural landmark. Two-thirds of the cost of the bridge was paid for by capital expenditures from the municipality, while the final third was paid for by the development corporation Nærheden Ltd.

The Hedeland recreational area is located adjacent to both Nærheden and Hedehusene. Hedeland has an outdoor amphitheater, ski run, golf course, riding schools, biking tracks, and many other amenities. This whole area was recently placed under a new strategic plan and may receive an overhaul in the years to come.



Image 11: Aerial view of Nærheden development area. Source: Google Maps, June 2020.

The western part of Nærheden was formerly an industrial area, while the eastern part was farmland. The municipality purchased the land in phases. Industrial facilities and warehouses were demolished in the years immediately before and after the land was handed over to Nærheden Ltd.

Sequencing

Nærheden's development is driven by a partnership between Realdania By & Byg and the Høje-Taastrup Municipality. The partnership, Nærheden Ltd., was established in 2013 with a team of three full-time employees. In the first two years, Nærheden Ltd. focused on developing its plans for the area. In 2015, Nærheden Ltd. began selling land assets. Construction began in 2017, and the first residents moved into the area in 2018. The entire development was initially scheduled to take

25 to 30 years to complete, but it now seems likely that the development will be finished within the next 8 to 10 years.

Nærheden will include an ambitious primary and lower-secondary new school focused on science. Nærheden Ltd. CEO Ole Møller says, “The municipality made a brave decision. It decided to close two rundown schools in the area and is building the new Science School to replace both” (interview, 27 January 2020). In total, 1,200 students will attend the school. The idea is that Hedehusene and Nærheden residents will be sharing the same school, helping to reinforce the desired socioeconomic diversity.

To achieve this, LEGO Education is partnering with Høje-Taastrup Municipality to develop the new school. The creation of a joint school supported by new opportunities for connectivity through the bridge and walking and biking paths represents one way Nærheden and Hedehusene are benefitting from their close proximity to one another. The school development is part of a wider initiative by the municipality to respond to Høje-Taastrup’s rapidly aging population by expanding the population of young families and youths.

The below timeline shows the sequence of the Nærheden development:

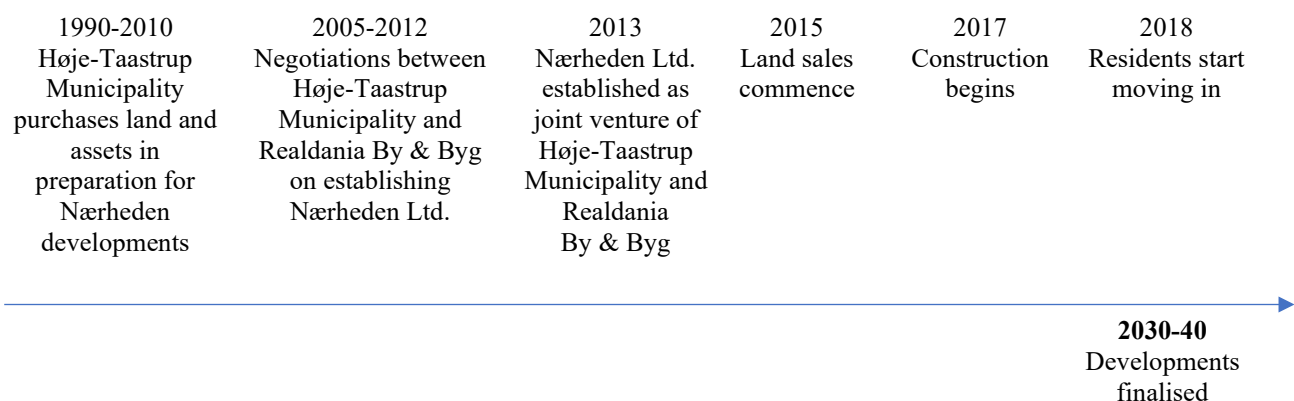


Image 12: Timeline of Nærheden developments. Source: Noring, 2021.

Although the Høje-Taastrup Municipality owns 50 percent of Nærheden Ltd., the development corporation is managed under a private regulatory framework. The corporation thus adheres to the legislation guiding private limited companies and is not required to conduct public tendering, public hearings, etc. The other 50 percent of the corporation is owned by Realdania By & Byg.

Strong Vision

The adjacent Hedehusene is a socially diverse suburban area that is characterised by single-family homes of the 1970s, but “it’s a bit old fashioned and dull,” CEO of Nærheden Ltd., Ole Møller, says (interview, 27 January 2020). The vision for Nærheden was to create a new area of density, proximity, and togetherness, but the owners could not be sure in advance whether there was market demand for this offering.

From the outset of development, Nærheden Ltd. was approached by three different investors that would have allowed Nærheden Ltd. to achieve a quick win and good profits. “But they wanted to build single-family houses, and that would have clashed with our core values of density, proximity, and togetherness,” says Møller (interview, 27 January 2020). Nærheden’s greatest challenge lies in balancing market demand and long-term benefit for the municipality as a whole. For this reason, when land is sold, price is not the decisive factor; rather, the new developments must foremost support Nærheden’s vision.

The market was initially slow to recognize the value of this vision, and Møller spent considerable time and energy “ringing doorbells, going to conferences, booking meetings with potential investors, etc. In the end, it paid off!” (interview, 27 January 2020). Once the development corporation had sold the first project—48 townhouses called Solrækkerne—to MT Højgaard in 2016, sales began picking up.



Image 13: Photo of Solrækkerne. Source: Martin Håkan/Coverganda.dk.

A principal finding of this working paper is that it is difficult to create a vision that is sufficiently different to stand out yet also sufficiently realistic to succeed. Nærheden’s development emphasizes intergenerational community living, offering a way for the municipality to cope with its aging population. The redevelopment project thus includes new forms of community-centered integrated living and leisure for both elderly individuals and families, with multiple generations living under the same roof.

Møller explains how developments like Nærheden must find a compromise between price and project: “If we only chose to adhere to our vision, we might never sell, and our projects might not be relevant to customers. If we only focus on price, we undermine the development project and in the long term, we jeopardize the entire development.” If the development corporation had compromised too much on its vision in order to attract its first investors, it could have risked hobbling this vision to such an extent as to cause the development to lose its uniqueness and long-term ability to attract residents and investors. The vision must be strong and unique, but it also needs to reflect and respond to market demand.

Møller highlights how the patient capital of Realdania By & Byg and Høje-Taastrup Municipality allowed the development corporation to reject initial investment offers and wait for partners that valued the project vision. The two partners possess overlapping objectives, but each must also deal with its own institutional pressures: for example, Realdania By & Byg desires long-term returns on investment and wishes the development to serve as a demonstration project for other cities, whereas the municipality is primarily focused on local impacts.

“PlusHouses” are a living concept developed by NREP and CASA Group, combining serviced elderly housing with family housing to create better intergenerational connections. Nærheden’s PlusHouses consist of 139 rental townhouses and apartments with a large communal area for eating and other social activities.

NREP is using its PlusHouses as an investment product for institutional and private investors. If successful, PlusHouses can be replicated elsewhere in Denmark and beyond. The PlusHouse concept offers relatively small housing units, in which multiple generations of the same family can live side-by-side. The housing is relatively inexpensive, and residents can purchase add-on services, including cleaning, cooking, childcare, communal activities such as yoga, elderly nursing care, and so on. Services are cheaper when purchased through the PlusHouse’s management company because the company is sourcing in bulk.

Another concept presented in Nærheden is driven by Bovieran Danmark, which is a joint venture between the Danish Sjælsø Management and Swedish property developer Fastighets AB Balder. The Bovieran concept builds upon the idea of having elderly housing connected to a large winter garden with a tropical climate year-round. The 55 cooperative housing units were sold out before they were completed. Bovieran was launched in Frederikssund, a municipality in eastern Denmark. The Nærheden development hosts the second Bovieran concept.



Image 14: Photo of Bovieran in Frederikssund. Source: Bovieran, 2020.



Image 15: The Bovieran complex seen from above, where the large glass construction hosts the wintergarden. Source: Bovieran, 2020.

Nærheden's development is centered around the concept of creating new and strong forms of communal living across social segments and age groups. A large survey by Kantar Group conducted on behalf of Realdania and knowledge center Bolius⁷ concludes that individuals in Denmark with strong neighborhood relations have a higher quality of life and feel safer and more

⁷ <https://www.bolius.dk>.

secure in their neighborhoods than residents with weak neighborhood relations according.⁸ Nærheden is fostering cohesive and high-quality communities.



Image 16: Rendering of the coliving areas of Nærheden. Source: Leif Tuxen, 2020.

Solrækkerne was the first development in Nærheden and consisted of prefabricated building modules (including interiors) assembled in advance. Only the facades are mounted onsite. The modules are locally produced in Lyngstykke by Scandi Build.

Organizational Structure

Nærheden Ltd.'s board of directors consists of the Mayor of Høje-Taastrup, Michael Ziegler, who is from the Conservative People's Party; the leader of the opposition party, the Social Democrats; two members appointed by Realdania; and one independent member, the chair, who is a former government minister and former chair of Copenhagen City & Port Development Corporation.

In addition to the CEO Ole Møller, who has had a long career in public and private positions, including head of the Transportation Council, municipal technical director, and municipal director, the company has two full-time employees. One person is in charge of managing the contractors for land preparations. She coordinates with the building investors and developers regarding when their

⁸ "New Investigation of Danish citizens and their Physical Environment" (2018).
<https://realdania.dk/nyheder/2018/12/ny-undersogelse-af-danskerne-og-deres-fysiske-rammer>.

developments can begin and how the land must be prepared for development. The other employee is tasked with master planning, communications, sales assistance, etc. Nærheden Ltd. also receives assistance from a highly professionalized team of technical, legal, financial, and other experts supplied by consultancies.

Ownership

Nærheden Ltd. was established in 2013 after several years of negotiations between Høje-Taastrup Municipality and Realdania By & Byg, which each own 50 percent of the development corporation. In 2015, Nærheden Ltd. began selling land assets and released its development plan, following a design competition involving three architecture firms. In response to the development plan, the municipality changed the local zoning and planning of the area in 2016. The development plan was revised in 2019, when it became clear that development needed to be less dense, as the target segments (young families and the elderly) were not receptive to high-rise living outside the city of Copenhagen itself. In turn, this transition to lower density caused Nærheden Ltd. to adjust its business case.

Nærheden Ltd. is prohibited from developing buildings itself due to its 50 percent public ownership. It is only allowed to prepare the land for further development, including decontamination of polluted soil, construction of road infrastructure, etc. Because the development corporation can neither construct buildings nor sell them on the market at a profit (including taking advantage of cheap land to sell them at below market rates), Nærheden Ltd. is obliged to achieve the highest possible price for the land it sells to private investors and developers—with the condition of course that investors and developers adhere to the overall development vision.

Investment

In 2013, Realdania By & Byg purchased half the land from the municipality. Realdania By & Byg and Høje-Taastrup Municipality then combined their land assets in the partnership, with each contributing 50 percent of the capital injection into the development corporation. The capital injection equaled the cost of the land, meaning that the municipality in practice simply contributed the land.

Ole Møller explains: “When people move from the city to the suburbs, they don’t want to live in a high-rise apartment. They want grass between their feet and access to green areas.” Therefore, the plans for high-density construction were scrapped, and the expected population density and return on investment were lowered. “The challenge is that we have the costs under total control. Everything is running according to plan. We’re adhering to the cost estimates and deadlines. But we changed the development plans and are building less densely without changing the revenue and profit targets,” Møller says.

Access to Capital

Nærheden is the latest in a series of urban redevelopments invested in by Realdania By & Byg, with others including projects in Køge, Fredericia, and Ringkøbing-Skjern. In each of these development projects, Realdania By & Byg and its municipal partner have developed a shared vision to guide the

positive development of small to midsize towns, some of which have experienced decades of depopulation and economic decline. Nærheden's vision is to develop a natural refuge near the city, experimenting with new forms of coliving across generational and socioeconomic divides. Realdania By & Byg works to prototype different types of successful urban development that can inspire other towns and cities, including municipal governments, consultancies, developers, and investors. Realdania By & Byg is particularly eager to create replicable and scalable examples, producing experience and solutions that can benefit other towns and cities engaged in similar efforts. Focus is placed on developing innovative methods, organizational systems, tools, development plans, quality management indicators, timelines, and resident dialogue practices, all of which can contribute to new pathways toward communal living, climate adaptation, traffic reduction, etc. Each of the developments in which Realdania By & Byg invests has a distinct vision and, if successful, could be adapted and adopted by other Danish towns and cities.

Realdania By & Byg's philanthropic investments in the four urban area development corporations considered here (i.e. Høje-Taastrup, Køge, Fredericia, and Ringkøbing-Skjern) should be understood as patient investment with long time horizons and modest expectations of returns. This reflects the broader aim of these investments that is to reveal new high-quality forms of living and working that are both inclusive and sustainable. Realdania By & Byg will hopefully recoup the entire invested sum, ideally with small but positive returns. This should occur to the extent that the individual development corporation's finances permit it, and the last of the capital can be released once the development project is complete, and the development corporation can be dissolved.

Lastly, pension funds, both Danish and foreign, are becoming increasingly important investors in urban redevelopment. Pension funds have traditionally invested in stocks and bonds, but as bond yields decline, pension funds are looking to invest in property as solid and stable long-term investments. With Danish mortgage providers offering low interests on 30-year mortgages, investments in Danish property development have become a suitable substitute for bonds. The challenge is that, as Ole Møller says, "the pension funds only want to rent out in order to have the long-term steady revenue, and we need a mix of housing, rental, ownership, cooperatives, affordable, etc." (interview, 27 January 2020).

Conclusion

Nærheden's core strength is the powerful vision guiding its developments. This vision is rooted in the municipality's desire to diversify its population and Realdania's desire to develop an innovative concept that will assist midsize Danish towns undergoing demographic and economic decline.

In short, from the perspective of Høje-Taastrup Municipality and Realdania, efforts to encourage greater socioeconomic diversity in the municipality are enhanced by the following:

1. The partners aim to increase the number of young families and young people in order to counter the rapidly aging population (with many residents of 80+ years). This is achieved through experimentation with new multigenerational forms of living, which invite young families to live in close proximity to older generations, and by providing a state-of-the-art "super school," incentivize young families to move to the area for good school access.
2. The partners aim to attract more affluent residents through providing a single school for all the children of Nærheden and Hedehusene (the aforementioned "super school") and developing

new and convenient forms of connectivity between the two neighborhoods (in addition to shared use of the regional train station).

Realdania By & Byg is permitted to own buildings and other assets and earn money, while Realdania Filantropi serves only charitable purposes and cannot act commercially. The collaboration between Realdania Filantropi and Realdania By & Byg allows both partners to invest ethically for the long term. Realdania is a crucial partner in providing patient capital and relevant know-how on urban redevelopment, supporting Nærheden's overarching vision. Høje-Taastrup Municipality crucially brings the land and zoning rights to the partnership and invests in a new ambitious school and other public facilities.

When complete, Nærheden will provide 3,000 housing units for 8,000 residents. Nærheden aims to increase the population of new and affluent residents, which will lift up the municipality as a whole. Nærheden's progress has economic, social, and environmental elements. Affluent residents will boost local commercial activity and add to the area's socioeconomic diversity.

Holstebro

Introduction

Kirsten and Jørgen Færch established the Færch Foundation in 1984. The Færch Foundation's vision is to contribute to the economic revitalization of Northwest Jutland, including the town of Holstebro, which is our next case study. The foundation's vision is also to serve the Færch family. Thus, this family foundation contains two branches; one that manages a large property portfolio and other investments that help fortify the wealth of the foundation, and another that manages a large endowment that is dedicated to local development in and around Holstebro, where the Færch family's wealth originally derived from the fishing industry but was later further amassed through the tobacco industry. In 1869, Rasmus Færch established a tobacco company, which, in 1961, became the Scandinavian Tobacco Company, when it merged with the tobacco businesses of two other Danish families (Oben and Augustinus). In 1969, the Færch family invested in the production of plastics, and in 2008, the Færch family sold its shares in the Scandinavian Tobacco Company. Finally, in 2014, the Færch family sold Færch Plastic to IKT Capital Fund. Thus, between 2008-2014, all business activities were seized, and all assets were liquidized, which led to the expansion of wealth and the formation of a new strategy of the Færch Foundation.

The Færch Foundation is involved in a number of important projects in Holstebro, including the redevelopment of the land belonging to a former hospital. When Herning's Regional Hospital was established in 2007, this freed up seven hectares of prime municipal-owned development land in Holstebro. Since mid-21st century, Denmark has experienced a change in both demography and family structures. This is, amongst others, due to an increasing number of elderlies and divorcees that has seen a rise in smaller households, where living in close knit communities can help both in terms of physical cords, providing a sense of security and socially. In addition, there is an increase in young single-person households that also seek new and other forms of community living. Subsequently, a desire and demand arose to move away from the development of stand-alone single-family homes, which dominate the urban landscape in and around Holstebro. Holstebro has a

population of 37,000. The foundation supports a wide range of activities, including entrepreneurship, business, culture, and education.



Image 17: Illustration of the disposition of important development areas in Holstebro. Source: The Town Plan: Holstebro's urban development plan for a green cultural town. March 2020.



Image 18: Grounds of the former hospital. Source: Luise Noring, February 2020.



Image 19: The former hospital. Source: Luise Noring, February 2020.



Image 20: Detail of the area planning of the former hospital grounds. Source: The Town Plan: Holstebro's urban development plan for a green cultural town, March 2020.

Sequencing

Holstebro Udvikling P/S, jointly owned by Holstebro Municipality and the Færch Foundation, ran a collaborative resident engagement process, which ultimately led to the submission of around a thousand proposals for the town's future development. A series of parallel competitions was used to both winnow and combine these innovative ideas from residents, resulting in 20 proposals that were

then integrated into the town's master plan. The process was finalized when the Danish architecture firm Vandkunsten won the master plan for Holstebro, with development beginning in 2019 and projected to reach completion in 2049. Holstebro's town plan⁹ presents a new vision for Holstebro's urban core and the seven hectares (70,000 m²) of former hospital grounds. The focus of all four development areas is to create a green urban environment, and the projected planting of 1,000 trees in the town center in 2020 will kickstart the entire development. Land for the trees is freed up through the elimination of street-level parking spaces and through the revitalization of existing neglected green areas.

Claus Omann, CEO of the Færch Foundation, says: "In our resident engagement, we focused on identifying the problems, not the solutions, because we need to understand the problems properly before we can determine the solutions. People go too fast into solution mode without having sufficient understanding of the problems" (interview, 19 February 2020). Ample time was thus dedicated to gathering and processing input from residents.

During the resident engagement process, particular attention was paid to engaging the town's youths, given that Holstebro had experienced outmigration of young people for decades. Holstebro Udvikling P/S felt it crucial to properly understand why young people were leaving Holstebro and other small and midsize Danish towns. Because young people did not traditionally participate in urban planning consultations, Holstebro Udvikling P/S deployed new measures to engage young people, including visiting local schools, knocking on doors, and speaking with residents in the streets.

Although the Færch Foundation recognizes that significant resident engagements make the master planning process both more complex and more time consuming, it believes that the town needs to reflect the desires of its residents in order to halt depopulation and economic decline. In addition, the foundation believes that extensive resident engagement is crucial to improving the town's quality and to easing the implementation process for development projects.

The below timeline illustrates the sequence of key events in Holstebro's urban revitalisation:

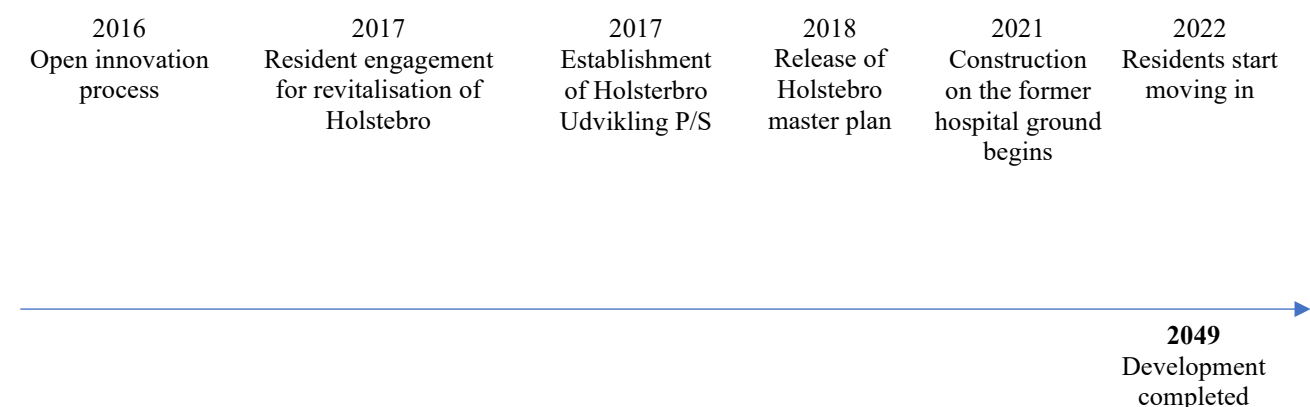


Image 21: Timeline of the Holstebro development. Source: Noring, 2021.

⁹ "The City Plan of Holstebro – A Green Trading City" (2020). <https://holstebroudvikling.dk/byens-plan/>.

Strong Vision

The vision is built around the desire to create more residential housing in the town center in order to foster resident interaction and communal activities. The resident engagement process helped shape Holstebro's long-term development, revealing new opportunities, such as the creation of a space that combined a boarding school, a museum, residential areas, and workplaces. Claus Omann says: "It's the coexistence that creates greater value than each of these can do on its own. It's a patchwork of activities. An ecosystem of economic and sustainable synergies" (interview, 19 February 2020).

The development plan's vision is to create a green cultural town, which the municipality and its development corporation will achieve through six initiatives: 1) increase the number and density of residents living in the town by building multi-family homes (as opposed to outside the town in one-family homes), 2) creating green areas (e.g. the planting of 1,000 trees in the town's core) with recreational activities, 3) creating livelier streets through an increased concentration of street-level commercial activities, 4) creating new connectivity across the town through a network of biking lanes and walking paths, 5) removing parked cars from the street by building centrally located parking facilities along one main street encircling the new area of development, and 6) bringing cultural activities out into the street through a close partnership with the Talent Academy.

The Færch Foundation and Realdania Filantropi donated DKK 20 million each to establish the Danish Talent Academy.¹⁰ The Danish Talent Academy, which opened in 2018, aims specifically to retain and invite young people to live and study in Holstebro. Each year, the academy gives around 40 students between the ages of 16 and 25, the opportunity to live on campus and study different arts and creative subjects. As student numbers continue to grow, so too does the number of young people living and studying in Holstebro—a remarkable accomplishment given the tendency for young people today to travel to Copenhagen to study.



Image 22: Source: Luise Noring, February 2020.

The Færch Foundation also recognizes opportunities for what it terms "mini-urbanization," namely the tendency for people over the age of 50 to move out of single-family houses in rural Denmark and into small and midsize Danish towns. For this reason, there is a focus on creating housing for elderly residents that are invited to participate and contribute to the green cultural town. Claus Omann and his colleague Hanne Torp Brodersen say that, in 50 years, everybody in Denmark will

¹⁰ The Talent Academy: <https://www.talentakademi.dk>.

be living in towns and cities, “which is better for our environment because we need less resources when we move together. We can create smart, new communal living” (interview, 19 February 2020). The Færch Foundation is aware that the economy is built around consumption, and it wishes to integrate new forms of self-sufficiency into Holstebro’s plans for smart communal living. Omann also recognizes the need for new urban solutions vis-à-vis climate change, saying metaphorically that “with flooding, we need to rethink the concept of houseboats” (interview, 19 February 2020).

Organizational Structure

The Færch family sits on the board of the Færch Foundation. Apart from family representatives, the board is comprised of highly specialist advisors.

Currently, Annemette Færch serves as chair of Holstebro Udvikling P/S, and Holstebro’s Social Democratic mayor, Hans Christian Østerby, serves as vice chair. The local leader of the opposition Conservative People’s Party also sits on the board. By having opposing parties represented on the board of directors, plans and proposals transcend local elections and political leadership, because no matter which party is in power, both opposing parties have embraced the plans and proposals decided by the board. In some instances, this may result in a longer period of time required for reaching unanimous board decisions. However, once decisions are made, they have to full support across the political spectrum, which allow them to be implemented more speedily and painlessly than would have been possible, if one party was opposed to the plans and proposals.

The board has two observers: Rasmus Færch and Lars Møller, the latter of whom is the municipal director. Omann says: “It’s important that I don’t sit on the board because I have to solve the problems that the board raises. Also, the municipal director is my counterpart, so I can’t sit on the board because that would distort our relationship” (interview, 19 February 2020). Omann has himself been a mayor for four years, and he regards this as an enormous advantage for his work at the Færch Foundation. He recognizes the importance of the foundation operating on a long timescale: a foundation can make bolder decisions and be more visionary than can either private investors or publicly elected officials.

Ownership

The Holstebro Udvikling P/S development corporation responsible for redeveloping the former hospital grounds and other areas in Holstebro is owned 51 percent by Færch Foundation and 49 percent by Holstebro Municipality. Both parties recognise the potential value of public-private-civic collaboration, yet such collaboration remains unusual in Denmark, where, up until around a decade ago, it was inconceivable for the public sector to collaborate with private or civic sector agents. Even today, a municipality cannot own and develop land, and public sector actors can only engage in urban development if they own less than 50 percent of a project. Thus, while a municipality can own and sell land, it is not allowed to develop that land and sell the land with buildings. This rule exists in order to make sure that the public sector actors do not distort markets by operating in the market on market terms, as the public sector actors possess the crucial advantage over the private investor and developer of conducting planning, zoning, permissions, financing and delivering infrastructure, etc. In this favorable position, public sector actors can distort markets, because they can sell at a sub-market price and still make money, as its’ risks and transaction costs are radically

reduced. In addition, the municipality may have attained the land at a favorable price, and in any event, it is spending taxpayers' money. As a result, it is the development corporation, rather than the municipality itself, that owns the land and creates the development plans for the land.

An alternative arrangement would have been for Holsterbro Udvikling P/S to simply purchase the land and develop it on its own. However, Omann is aware of the challenges involved in such a setup: "We're very conscious of not becoming illegitimate partners because if we coown the development corporation with the municipality that owns the land, other private investors may think that we're getting preferred treatment and access to the best land and the best deals. This presents a risk to our reputation, which we can best eliminate by inviting private investors to join the development corporation" (interview, 19 February 2020).

Investment

The Færch Foundation partnered with Holstebro Municipality to form the Holstebro Udvikling P/S development corporation. The development corporation was created in 2016 with DKK 8 million from the Færch Foundation and DKK 8 million from the municipality. The Færch Foundation primarily engages in impact investments, which the foundation interprets as doing good while making good money. The revitalization of Holstebro, including redevelopment of the former hospital grounds, falls within this category. Omann explains the transition that the foundation is undergoing in the process: "We're changing from becoming the invisible benefactor to becoming the visible societal agent" (interview, 19 February 2020). The foundation seeks to create a sustainable and future-oriented development plan for the town as a whole, as is evident from the forward-looking master plan of 2020.

Access to Finance

The former hospital grounds comprise 10 building plots on seven hectares. The construction is estimated to cost DKK 1.5 billion, and the development corporation is targeting large Danish pension funds as the principal investors. Danish pension funds mainly invest in Copenhagen but are increasingly seeking investments elsewhere in the country because Copenhagen's property market combines high prices with low interest rates, making it difficult for pension funds to achieve the desired yields. Nevertheless, when Danish pension funds invest in small and mid-sized Danish towns, they prefer to invest with local actors that possess local knowledge of the market and know how to add value to the investments. Onboarding local investors also provides the joint buy-in that the pension funds require for securing future investments. In short, local investors validate local investments.

Conclusion

In leveraging its future potential, the development of Holstebro takes a holistic approach. First, this is achievable through the sheer scale of development vis-à-vis the size of the town itself. In comparison to the size of the town, the development areas are substantial. Second, the areas under development are centrally located in the town and include the town center. Finally, the development plan presents six initiatives that in their entirety touch all areas of urban life and work, including mobility, consumption, and recreational activities. This makes the plan all-encompassing,

ambitious, and daring. It is possible for the relatively small town of Holstebro to achieve this plan through considerable institutional and financial capacity represented by the Færch Foundation and Realdania Filantropi, which each in their own way help the municipality realize the town's potential as a green and cultural hub of Northwest Jutland. Unlike many philanthropic foundations around the world, both the Færch Foundation and Realdania Filantropi have transitioned from only informing advising on policies and initiatives to actually carrying out the concepts they recommend. This makes such foundations more vulnerable to performance and impact measurements and subsequently vulnerable to public scrutiny, but more importantly, it enables transformative change emblematic of the urban development in Holstebro.

Fredericia

Introduction

This case concerns the development of Kanalbyen in Fredericia. The land stems from two different sources. One source of land was formerly the site of a chemical factory, Kemira, which closed in 2004. This land is located east of the town center and west of the old harbor. In 2008, the municipality purchased the land belonging to the Kemira factory from Yara. The remaining land area is the former shipyard. Shipbuilding activities ceased in 2008.¹¹

The industrial harbor presented issues involving noise pollution, which required mitigation as the harbor remains operational. Noise and soil pollution presented the greatest obstacles to further urban development in the area. By the time the 20-hectare redevelopment zone is completed in 2040, it will host 1,200 apartments, 1,800 workspaces, 80,000 m² of office space, and 15,000 m² of retail space. Kanalbyen is emblematic of a remarkable transition from an environmentally damaging industry to transformative urban expansion.

Sequencing

First, the chemical factory, Kemira, closed in 2004, while the ship construction facilities closed in 2008. The municipality purchased the Kemira land in 2008. Fredericia's wider planning process was finalized in 2007, and the municipal plan was approved by the municipal council the following year. The municipal plan was a prerequisite for establishing and guiding the work of Kanalbyen i Fredericia P/S, the development corporation tasked with developing Kanalbyen. Kanalbyen i Fredericia P/S was set up in 2009, with 75 percent ownership by Realdania and 25 percent ownership by the municipality.

In that same year, Kanalbyen i Fredericia P/S managed and finalized an architecture competition, which was won by KCAP and Vandkunsten. The architecture competition set the detailed framework for Kanalbyen's development plan, in line with the wider municipal plan that was approved in 2008.

A neighborhood development plan released in 2012 offered Kanalbyen i Fredericia P/S the opportunity to request rezoning from the municipality, a process that concluded in 2013. This

¹¹ The official website of Kanalbyen i Fredericia P/S: <https://kanalbyen.dk/english/>.

allowed the development corporation to embark upon its local development plan in 2014, which provides the detailed framework for land sales to investors and developers. Individual investors and developers must apply for a building permission as the final step in the planning process. Construction of Kanalbyen began in 2017.

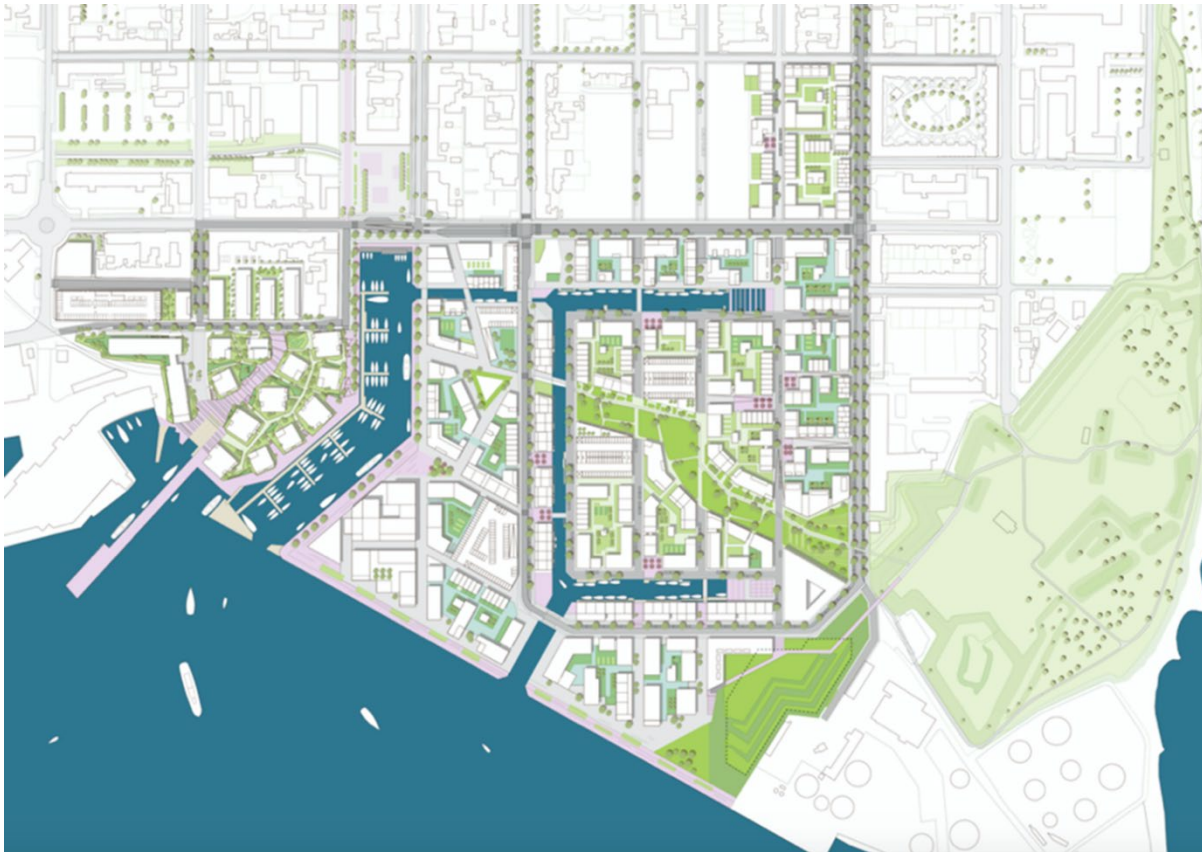


Image 23: Kanalbyen i Fredericia. Source: Realdania By & Byg, 2020.

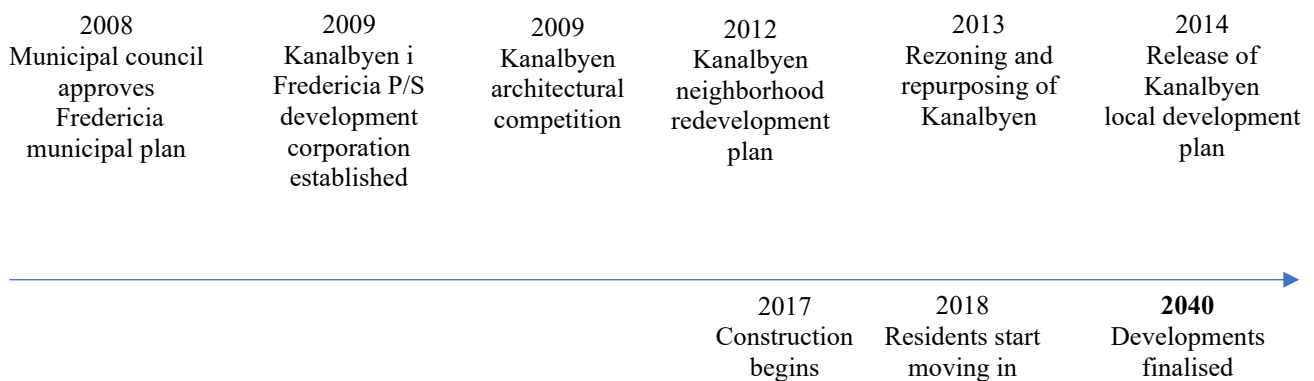


Image 24: Timeline of Kanalbyen i Fredericia. Source: Noring, 2021.

Strong Vision

Kanalbyen's marina and waterfront location mean that the neighborhood's vision builds upon the presence of the sea and the abundant natural beauty surrounding Fredericia. The Kanalbyen vision is a Blue Vision, which is unique for Fredericia. The vision centers on the activities and attributes accentuated by the proximity of the ocean. Kanalbyen has created a new connection between the old town and the sea. The town proper is home to an old fortress and town center dating back to 1650. Prior to Kanalbyen's construction, however, access to the sea from the town proper, including from the fortress and old town, was blocked by heavy industry and the shipyard. The new connectivity and townscape make the entirety of Fredericia more attractive to locals and visitors alike. To reinforce this vision, Kanalbyen will create three canals with a publicly accessible marina and plenty of sea sport activities.

In partnership with the municipality, Realdania also invested in measures to combat persistent stormwater flooding and to make the waterfront more resilient to rising sea levels. This is evident in the below picture, which shows how the land is raised considerably above sea level.



Image 25: Kanalbyen i Fredericia. Source: Joan Barløse, 2020.

Tim Halvorsen, CEO of Kanalbyen i Fredericia P/S, says: “We changed the development plan, but we never changed the vision. The vision must remain intact, while strategies and tactics can change” (interview, 10 June 2020). One example is the mini-academies that were originally planned for Kanalbyen. Fredericia had hosted a nationally renowned theatre until the COVID-19 crisis broke out, and the theatre was forced to close. The idea was to build a mini-academy for the theatre

arts, but with the theatre closed, those plans were abandoned. In much the same way, a local school is planned for the area, but the project has been postponed until there are sufficient children to attend the school.

Taking inspiration from Fredericia's tradition of expressive art and culture, art and culture are key components of Kanalbyen's urban development vision. Kanalbyen's website states: "Kanalbyen works strategically with art as a tool to create life, engagement, meetings, identity and cohesion."¹² Art installations are part of the very early developments and help create the connective tissue across the entire development area. Throughout the area, art displays are located so as to tell a story of the town's history as a hub for fishery and industry.



Image 26: Works by the artist Gunhild Aaberg. Source: Kanalbyen i Fredericia, 2020.

In 2018, the National Art Foundation (*Statens Kunstfond*), which is a public institution, granted funding for the creation of integrated artworks across six large building constructions in Kanalbyen. During 2019–2020, the first two art installations were completed, one of which is illustrated in image 27.

¹² <https://kanalbyen.dk/oplev-kanalbyen/kunst-i-kanalbyen/>.



Image 27: Works by artist Signe Guttormsen nævnes. Source: Jan Søndergaard, 2020.

Fredericia is part of Denmark's Triangle Region, which is located on both the Jutland Peninsula and the island of Fyn and is home to around 422,000 people.¹³ The Triangle Region encompasses important aspects of Danish industry, such as fashion and textiles, wind turbine manufacturers, and industrial pump manufacturers (for district energy).

The challenge is that the major towns in the Triangle Region (Kolding, Vejle, Fredericia, etc.) are all competing with one another for residents and investment. For instance, Halvorsen explains, Kanalbyen is competing with all the other cities in the Triangle Region, which are undergoing similar major urban development processes, including Marine City in Kolding.

Fredericia's Kanalbyen is directly adjacent and connected by foot to the fortress and old town center. Halvorsen explains that Kanalbyen not only grants the opportunity to build at close

¹³ <https://www.trekantomraadet.dk>.

proximity to the fortress and old town but also carries responsibilities as result of its location. Each development project in the Triangle Region needs to stand out with a clear and strong vision in order to compete with other mid-sized towns in the region. Mixed messages drown in this market. “There’s extreme competition in the region,” says Halvorsen: “We need to be clear about what we offer that’s distinct from other offerings in the region” (interview, 10 June 2020). Kanalbyen’s distinctiveness is expressed in its sea, arts, and culture vision.

Organizational Structure

The Kanalbyen i Fredericia P/S development corporation is a joint partnership between Realdania and Fredericia Municipality. Tim Halvorsen is its CEO, and it has three full-time employees. The corporation outsources much of its specialized work to temporary project employees. The board of directors consists of five members, including the chair Carsten Koch, two board members from Realdania, and two from the municipality.

Ownership

Kanalbyen i Fredericia is 75 percent owned by Realdania and 25 percent owned by Fredericia Municipality. Realdania’s motivation for embarking on the Kanalbyen redevelopment and other similar urban development projects is to demonstrate how to develop high-quality towns and cities in markets that cannot carry out the developments themselves. Halvorsen says that Realdania “has enormous competencies. We built on their project competencies and put developments into a 30-year timeframe. It’s a combination between long-term urban development and short-term project funding” (interview, 10 June 2020).

Halvorsen continues, “the towns were chosen [for urban development] because Realdania believes that with the right mix of competencies and investment, these towns will be able to stand on their own two feet” (interview, 10 June 2020). Peter Cederfeld, CEO of Realdania By & Byg, is the main shareholder of the four Realdania coowned development corporations presented in this working paper. Peter Cederfeld also sits on the board of directors of all four development corporations as one of the Realdania board representatives.

Investment

There are 60 building plots in Kanalbyen. The price of the building plots is confidential and subject to negotiations based on market trends. “The price is our management tool. We know the market, and we know what our land is worth,” Halvorsen says (interview, 10 June 2020). Investors and developers investigate the town’s historic population growth in order to determine the market potential and set the land price. Fredericia has experienced modest population and economic growth for the past decade. Kanalbyen is targeting a middle segment of the market.

The investors and developers also take into account what projects have already been constructed or are in planning. For example, if a large residential area aimed at housing for the elderly has been planned, then the investor or developer will seek to build something else.

Access to Finance

The investors are often large Danish pension funds, such as AP Ejendomme; large private nonprofit housing cooperatives, such as Boligkontoret Fredericia, which owns 6,000 rental apartments locally; and local and regional investors and developers, such as O. Adsbøll, which has joined forces with the housing cooperative Domea to leverage investment capital through socioeconomically mixed residential development.

Danish pension funds are primarily interested in long-term investments and therefore prefer rentals that they can own for 30 to 40 years. This is also the case for private nonprofit housing cooperatives, which own affordable rental housing and deploy a revolving funds model, in which all revenue is reinvested into the maintenance of existing housing and construction of new housing. The shared long-term investments support collaboration between pension funds and private nonprofit housing cooperatives. In contrast, private investors and developers prefer shorter term investments and thus favor owner-occupied residential units that can be sold off once finished. The result is a mixed neighborhood spanning different socioeconomic segments. Finally, the publicly accessible marina is a municipal investment funded by taxpayers' money.

As we have already noted on several occasions, timing is crucial when it comes to these kinds of development projects. In its first years, Kanalbyen sold few plots of land. To keep afloat, the development corporation required a capital injection of DKK 70 million from its owners (proportionate to their shares of investment), which it used to build the first canal, the trademark of Kanalbyen. "We had to show we were serious. That we meant it. We had to move ahead with the development," Halvorsen says (interview, 10 June 2020). This was the first of three canals was completed in 2014–2015, near the end of the financial crisis in Denmark. A capital injection of DKK 70 million prolonged the return on investment period. It also made the corporation more vulnerable to further market fluctuations, as performance targets were tightened.

Conclusion

Realdania By & Byg has been instrumental in transforming the former industrial town of Fredericia, which once had a busy shipyard and heavy industry that shut off the town from the sea. Kanalbyen is reconnecting Fredericia with its maritime roots through the artistic expression found in the integrated works of art throughout Kanalbyen.

Kanalbyen i Fredericia P/S is entrusted with the financial tools to translate the town's ambitious urban quality vision into completed projects. Due to its revolving funds model, Kanalbyen i Fredericia P/S development corporation need not prioritize maximizing income from the sale of each individual parcel but can instead work to foster new and unique markets; developers are in turn incentivized to compete in strategic areas that increase overall, long-term quality, rather than to make bids focused on price. In order to successfully navigate this strategy, Kanalbyen i Fredericia P/S development corporation creates synergies between stakeholders and results in new developments of high urban quality. Other places can follow Fredericia's approach by working with private developers to create targeted, catalytic projects. Examples from Fredericia include projects accentuating the close presence of the sea and integrated installations of arts that can reshape and redefine neighborhoods and cities throughout their redevelopment.

Ringkøbing-Skjern

Introduction

Ringkøbing-Skjern is the largest municipality in Denmark in terms of landmass, encompassing five small towns and 40 villages. The economy relies on manufacturing, tourism, and public sector employment. Mayor Hans Østergaard explains, “Towns don’t compete against each other [within the municipality]. We let the development happen where it naturally occurs. I’m not on a mission to make people move from the villages to town. We see that a lot of the villages have a big drive to develop and offer something new.” However, the mayor also notes a tendency toward “internal urbanization,” in which the municipality’s towns and villages attract residents from its rural areas.

Ringkøbing-Skjern’s development project is called Naturbydelen. Naturbydelen is located on former agricultural land, which the municipality purchased from a farmer many years ago. The area is located on the shore of Ringkøbing Fjord, close to the North Sea and the Holmsland Dunes, an area of stunning natural beauty. Naturbydelen is located 500 meters from Ringkøbing town center, which offers an abundance of small shops and busy streetlife. The draw of the town itself is an important sales proposition for Naturbydelen.

The entire development is 80 hectares, of which 60 hectares are set aside as a managed nature zone, and 20 hectares are dedicated to residential development. Naturbydelen is mono-functional in that it is intended solely for residential purposes. As a result, the development will not include summer houses, despite the attractive natural setting, or commercial or retail activities.

Ringkøbing has not experienced population or economic growth for nearly a decade. However, it has one of the lowest unemployment rates in Denmark and boasts a population with a relatively high level of education.

As a result of Ringkøbing’s stagnant population and economy, Naturbydelen is perceived as an experimental endeavor. Østergaard confirmed that Realdania sees Naturbydelen as an experiment and a potential demonstration project for replication and rescaling to other Danish towns. Tried-and-tested models for urban economic development offer other small and midsize towns a path forward.

Sequencing

After substantial competition, landscape architects SLA won the competition, and Naturbydelen’s development plan was released in 2014. In 2016, the natural landscape of Naturbydelen was established. Sales have progressed gradually since the first sale of two buildings (containing a total of eight apartments) in 2016. “We received a lot of valuable feedback on the early construction,” says Østergaard, “including that people did not want two-story houses, and they wanted their own outdoors activity shed for recreational activities, gardening equipment, and other possessions” (interview, 24 June 2020).

Sales of land plots in Naturbydelen started in 2018. To date, Naturbydelen has sold land for a total of 70 residential units of which 40 units were sold to Ringkøbing-Skjern Boligforening (an

affordable and social housing cooperative that is membership owned) who dedicated its development to family living, 14 units were sold to Realdania By & Byg and dedicated senior living, and 16 units were purchased by local private developer Frifeldt and are targeted privately owned residential living. In total, by the end of its 35-year development period, Naturbydelen will contain 1,500 residential units on its 20-hectare development area.

2009	2014	2016	2018	2020	2050
Establishment of Ringkøbing K Aps development corporation	Launch of Naturbydelen development plan	Natural landscape established	First resident moves in	20-30 new residents	Developments finalised

Image 28: Timeline of Naturbydelen development. Source: Noring, 2021.

According to Østergaard, after a slow start, progress on Naturbydelen has now reached a satisfactory pace. Nevertheless, he says, they had hoped for more sales by this point. Østergaard explains that the market has taken a long time to accept and embrace the Naturbydelen's unusual concept.

Strong Vision

The vision for Naturbydelen is to create new types of communal living, which draw on benefits from the beautiful natural setting. In the process, the developers have renovated a stream, created a vast meadow, and initiated a process of reforestation. Further, small flower and berry patches have been created, inviting residents to engage in their own communal gardening, and make their own juice in communal workshops. 'Weeds' are cherished as part and parcel of the setting's biodiversity. Naturbydelen's innovative orientation toward the private sphere and communal living appeals to people, who are interested in new and sustainable lifestyles in close synergy with nature.



Image 29: The Development Plan “Ringkøbing K – In the Middle of Nature”. Source: SLA, 2020.

The redevelopment zone does not contain any individually owned land. Instead, all residents have access to vast common areas that they can cultivate and form however they wish. Naturbydelen is in fact an enormous community garden. The land is managed by the community representative organization, which has hired a resident-financed professional gardening company.

Østergaard says, “We wanted to access new and interesting urban living opportunities. It’s a new way to build. There are requirements on the materials that we use. The houses must be zero energy. And there are very specific requirements concerning the outdoors environment” (interview, 10 June 2020).

Tim Halvorsen, CEO of Ringkøbing K Aps, explains the challenge faced by Naturbydelen: “We know how the ‘typical Dane’ wants to live.... in a single-family home” (interview, 10 June 2020). In Naturbydelen, residents are invited to live in shared housing and collective communities and participate in other forms of cohabitation.



Image 30: The Strandkanten project. Source: Søren Palmelund, 2020.

The developer has targeted a wide array of populations, not just those interested in trying out the new forms of coliving offered by Naturbydelen. The developer has also targeted people interested in convenient living, as the housing is new and built with convenience in mind, and the garden requires no maintenance. Naturbydelen also targets people drawn to the outdoors, who might appreciate the development's opportunities for activities such as kitesurfing, biking, running, sailing, fishing, and hunting.

The development corporation sells a parcel of land as a fully developed project. Thus, the corporation does not sell a plot of land without knowing what the final project will look like. For instance, the corporation does not sell individual plots of land for individual single-family homes. Also, Naturbydelen is being developed from the core outward in order to ensure the desired townscape from the outset. Throughout the course of development, development corporation determines where buildings will be positioned, and where it is necessary to retain expanses of pure nature. Hans Østergaard explains that the municipality does not conduct tendering but rather waits for potential investors and developers to approach the municipality, after which a one-to-one dialogue between the municipality and an investor or developer begins.



Image 31: Senior citizens housing. Source: Søren Palmelund, 2020.

Most new residents come from West Jutland, but Naturbydelen has also been able to attract new residents from other parts of the country, including Copenhagen. Østergaard points out that Naturbydelen offers many rental units because renting frees up time in terms of maintenance and frees up money for spending, especially for retired residents who sell their properties elsewhere. Last but not least, renting gives residents the flexibility to easily move if they get a job elsewhere.

Organizational Structure

The development corporation Ringkøbing K Aps is owned 75 percent by Realdania By & Byg and 25 percent by Ringkøbing-Skjern Municipality. In addition to CEO Tim Halvorsen, Ringkøbing K Aps employs Senior Consultant Ann Hein. The corporation is tasked with selling plots of land to investors and developers, preparing the land for construction, and marketing the community. Ringkøbing K also leads Naturbydelen's neighborhood redevelopment plan.

The corporation has an independent board of directors, of which the municipality appoints two members and Realdania appoints three, including the chair. The municipal council is responsible for municipal and local planning and building permission.

Ownership

In 2011, Realdania By & Byg entered into a partnership with Ringkøbing-Skjern Municipality to build Naturbydelen. Tim Halvorsen is CEO of both Ringkøbing K Aps in Naturbydelen and Kanalbyen i Fredericia P/S in Kanalbyen (discussed earlier in this paper).

Investment

Naturbydelen offers private nonprofit housing projects to investors, ranging in price from DKK 4–5 million. Typical investors are regional and local housing organizations, such as Domea, By Midtvest, and Ringkøbing-Skjern Housing Association. The price of private nonprofit housing is DKK 7,000–8,000 per month for a 100–120 m² apartment. In line with national law, the municipality contributes 10 percent loans to private nonprofit housing associations, which are repayable over a 40-year period. In addition, the municipality often offers discounts on the land price. This is to counter a longstanding tradition of private investors purchasing and renovating older buildings in the towns of Ringkøbing-Skjern. “When no private investors want to build, the municipality has to take the initiative,” Hans Østergaard says (interview, 24 June 2020). However, Østergaard points out that the land and construction costs in Naturbydelen are cheaper than in Copenhagen. The development corporation and municipality value the know-how that the regional and local private nonprofit housing associations bring.

Access to Finance

There are two types of investors: 1) regional and local investors, such as Frifeldt, that have decades of experience building summer houses; and 2) private nonprofit housing organizations, such as a housing association, Ringkøbing-Skjern Boligforening, specializing in zero-energy housing. Additionally, Realdania By & Byg is using Naturbydelen as a site for experimentation and built 14 homes for the elderly for private rental in 2020, which were subsequently sold to a private investor.

Conclusion

Like many rural municipalities, the towns and villages of Ringkøbing-Skjern are struggling with depopulation and economic stagnation. This project seeks to make the most of the development area’s close proximity to nature and the sea by providing dense, high-quality living opportunities for families with children as well as other demographics that wish to live close to nature. Over the next 30 years, 1,000 housing units will be constructed. Prior to starting construction, however, the whole area was repurposed and re-landscaped, creating a large recreational meadow with forest, lakes, berry patches, and walking and biking paths. Nature is at the center of this mono-functional development. In some ways, the singular focus of this development makes it both highly experimental and daring, and it offers a way to compete and differentiate itself in a market that emphasizes multifunctional and mixed living/working urban areas.

Aalborg

Introduction

With a 2017 population of 211,000, Aalborg is the third-largest city in Denmark, after Copenhagen and Aarhus. Aalborg is located in the northern part of the Jutland Peninsula. Historically, Aalborg was a highly industrialized city with thriving shipyards, cement, grain, and fodder production industries. Aalborg Municipality is at the helm of three large redevelopment projects: 1) the central waterfront on the fjord's Southern side, which is developed mainly by the municipality itself, 2) Aalborg East, which is largely developed in collaboration with affordable and social housing associations, and 3) Stigsborg Havnefront on the northern side of the fjord, which is the case presented in this working paper. This development project is particularly interesting because it presents an entirely new form of governance and financing of area development. The full development of Stigsborg Havnefront is estimated to take up to 25–30 years. It will become the largest “city park” in Aalborg, a city that has experienced population growth since 2006.

Infrastructure investments are part of the “value adding package” offered by Aalborg Municipality. Because Aalborg Municipality recognizes the necessity of good infrastructure for global market opportunities and business growth, it maintains investment in an internationally connected harbor and airport. The port's annual production output is nearly US\$2 billion, with an employment effect of almost 10,000 jobs. Since 2007, the port's production output has increased by 54 percent, and the employment effect has increased by 44.5 percent. The shift to entirely new, cutting-edge harbor facilities sought to simultaneously keep the harbor competitive and free up land for urban expansion within the city proper. With over 1.5 million passengers annually, Aalborg Airport is the third – largest airport in Denmark. Since 2006, international destinations from Aalborg Airport have risen from eight to 40.

Aalborg Municipality, PFA Property A/S, and A. Enggaard A/S entered into a partnership in 2018 to form the Stigsborg P/S development corporation, which is tasked with repurposing the deindustrialized land and conducting urban regeneration at Stigsborg Havnefront. PFA Property A/S is the property investment and development arm of the private sector cooperative PFA Pension. Aalborg Municipality owns 49 percent of the Stigsborg P/S development corporation, and the private actors PFA Pensions and A. Enggaard 51 percent.

Stigsborg Havnefront, site of a late-Middle Ages fortress, is located along the shores of the Limfjord, adjacent to the bridge to Aalborg's city center. Aalborg Municipality purchased the land from private industry owners in the 1990s, and the city accumulated land along four axes projecting out from the city center. As a result, the city now owns much of the land discussed here, though some of it has since been sold to Stigsborg P/S. In 2012, the city released a master plan for developments along the four axes. The municipality is only a major landowner in the Stigsborg Havnefront and Centrale Havnefront developments; the other development land is predominantly privately-owned.

At 55 hectares, Stigsborg Havnefront is among the largest single urban redevelopment projects in Denmark. Construction began in 2018 and is scheduled to finish in 2045, with an expected 4,000 residential units and 7,500 residents. Ninety percent of the planned 400,000 m² in floor area is

expected to be residential space. The remainder will consist of amenities, shops, institutions, schools, and office space.

The developments are scheduled to occur in three phases. The first phase, which involves 16 hectares of land formerly owned by the municipality, will be completed in 2028. Over the span of two years (from 2018 to 2020), Stigsborg P/S sold 25 percent of the land in phase one, the equivalent of 50,000 building plots. Phase two consists of 11 hectares, which are owned partly by Aalborg Port and partly by the municipality. The final phase, consisting of seven hectares, involves land owned by the municipality. The remaining land (16 hectares) will form a publicly funded park. There will also be a publicly accessible marina for small boats as well as a crossing over the fjord, either in the form of a pedestrian and bike bridge (partly financed by Aalborg Municipality, Stigsborg P/S, and philanthropic sources) or, if the finances prove too difficult, in the form of a water bus. The Stigsborg P/S development corporation is entitled to purchase land in all three phases of development.



Image 32: Aerial visual of the development area prior to development. Courtesy: Realdania By & Byg, 2020.



Image 33: Aerial visual of the development area prior to development. Courtesy: Realdania By & Byg, 2020.

The development plan¹⁴ ensures that as many housing units as possible benefit from a view of either the water or the park. Population density is thus highest near the water and the park. All buildings are between two and six stories tall.

The timeline in image 36 illustrates the sequence of events of developments in Aalborg Stigsborg Havnefront:

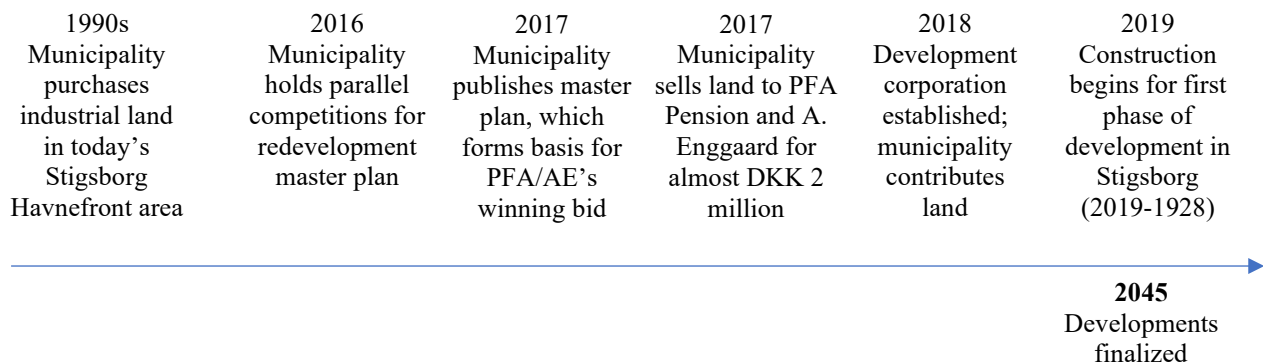


Image 36: Timeline of the Aalborg development. Source: Noring, 2021.

Strong Vision

“Stigsborg Havnefront creates synergy and integration with Aalborg city center. This new area provides optimal conditions for life in the city, on and by the water, for families and companies and for people wanting to use and experience the area's unique character. Stigsborg Havnefront wants to create a growth center for the region and the nation, while continuing to build strong international relations.”¹⁵

“Smart” solutions are integrated into the city building plans. These solutions facilitate environmentally, socially, and economically sustainable ways of living, including a car-sharing arrangement, centrally located car parks (allowing large areas of the development to be largely car-free), high levels of biodiversity, and shared living and consumption arrangements. Finally, to pre-empt climate disruptions, the entire area has been elevated for protection from flooding and stormwater.

Yet, people do not simply move into neighborhoods; they move into cities. Stigsborg Havnefront and other large-scale urban redevelopments cannot stand alone. Such districts belong to the broader contexts of the cities that host them. Privately-owned areas can adopt the holistic contextual principals of Danish development corporations without privately-owned development corporations having jurisdiction over the entire city. For instance, privately-owned development corporations can implement renewable energy in the areas under development, provide largely car-free areas, offer expansive networks of bike lanes, and subsequently work with city leaders to expand these ideas

¹⁴ Udviklingsplan for 1. etape af Stigsborg Havnefront, Tegnestuen Vandkunsten et al., February 2017.

¹⁵ Freely translated from Stigsborg Havnefront website: <https://stigsborghavnefront.dk/om-stigsborg-havnefront/udviklingsplan-1.-etape> and the ‘Value- and Quality Catalogue for Stigsborg Havnefront’, March 2015.

and solutions beyond the areas under development. Nevertheless, this approach requires developments of substantial scale, which enable feasible returns on investment and influence the entire host city. In this way, distinct developments can propel investments across wider city contexts by encouraging city leaders and other private investors to follow suit.

In much the same manner, Aalborg's redevelopment of Stigsborg Havnefront is not a standalone project. It is the culmination of years of redefining Aalborg's economy and purpose. A sequence of projects has excelled in demonstrating urban quality and value. Such projects include the Utzon Center for architecture and design; House of Music, designed by the internationally renowned architecture firm Coop Himmelblau; Nordkraft,¹⁶ a former power plant converted into a cultural and music facility; and the housing cooperative designed by Henning Larsen Architects, which hosts 300 living units for young people. Finally, the derelict former freight train station was converted into an educational center cohousing several universities and other institutions.

Although these urban redevelopment projects did not focus on business development, they have contributed to a flourishing of start-ups and significant growth in jobs. The number of new start-ups per year grew from 1,300 in 2012 to just under 2,000 in 2016, resulting in the creation of 7,500 new jobs. Danish municipalities do not typically get directly involved in business development, but by creating attractive cities, they attract new residents and are better able to retain university graduates. This in turn expands the city's talent pool and skills capacity, attracting companies and incentivizing start-ups.¹⁷

To conclude, the City Architect Peder Baltzer Nielsen reflects on the transformation that the city has undergone, reasoning that Aalborg has shifted from an industrial city to a knowledge and service city, or rather knowledge and culture city. Baltzer Nielsen puts this in other words, saying that Aalborg transformed from a trading city to an industry city from 1900 to 1970. According to Nielsen, since 1970, the city has increasingly focused on the people living in the city and their quality of life. For this purpose, the city strives for great diversity and social integration, including the construction of 7,000 affordable youth and student apartments along the Southern waterfront.¹⁸

Organizational Structure

Anton Hessellund is CEO of the Stigsborg P/S development corporation, a partnership of Aalborg Municipality, PFA Pensions, and A. Enggaard. Each partner in this joint venture plays a crucial role.

The Municipality

The municipality contributes land for the project; develops public infrastructure (including a park, school, and kindergarten); and enables developments through planning, zoning, and permissions. Because the municipality's primary interest lies in the redeveloped area as such, rather than in profiting from the redevelopment, it is also eager to learn from its partners about how successful urban redevelopment of deindustrialized areas can be achieved at a very granular level. To

¹⁶ <https://nordkraft.dk/forside.aspx>.

¹⁷ Aalborg – Tough Little Big City, <https://www.aalborg.dk/media/6370314/tough-little-24siderfinal-small.pdf>

¹⁸ Peder Baltzer Nielsen, Byplanhistorisk Udvalgs seminar 2019.

maximize learning, the city has dedicated key municipal employees to the secretariat tasked with overseeing the redevelopment. The overall goal is to forego on profiting from the Stigsborg redevelopment in order to capture learning and experience that will allow municipality to conduct future redevelopment itself. Prior to the Stigsborg Havnefront development, the municipality had conducted tendering for developments along its four axes, and, as a result, the municipality did not undertake the actual development work.

Hessellund says that there are both challenges and opportunities to working with the municipality: “We work at a faster pace in the private sector. The city wants to know what’s going on and wants to stay updated. But we’ve also learned a lot about the city’s different approval steps” (interview, 26 June). Hessellund notes that the municipality has arduous reporting requirements and complex decision-making hierarchies (included decision making documents and consultation processes), in contrast to A. Enggaard’s flat organizational structure. “The city probably thinks of us as the impatient kid in the class who constantly pushes ahead. We all agree on the vision and want to create something of lasting quality. But PFA and Enggaard basically need to earn money” (interview, 26 June 2020).

Pension Funds

Like many Danish pension funds, PFA Pension is a major capital investor in towns and cities across Denmark. PFA Pension regards urban development as a preferred investment area. Danish pension funds primarily invest in rental housing, which provide stable revenues over a 30-year period. Rental prices in Aalborg are DKK 1,200/m² compared with DKK 1,800/m² in Copenhagen. “A couple of years ago, we couldn’t sell rental projects because nobody wanted to rent in Aalborg,” says the Head of Department, Anton Hessellund, who oversees developments on behalf of the property developer A. Enggaard (interview, 26 June 2020). Importantly, with new rental development projects, Aalborg has been able to tap the resources of institutional pension fund investors.

Danish pension funds take a portfolio approach, investing 80 percent of their funds in long-term 3–4 percent return on investment projects and 20 percent in what are perceived to be more strategic and risky projects with higher returns on investment. Thus, when Danish pension funds invest in offshore wind energy, these projects are considered strategic investments. Danish pension funds also continuously invest in the Danish renewable energy transition in close alignment with the Danish government’s green agenda and climate goals. Danish pension fund investments in rental housing are, in contrast, perceived as long-term, low-risk investments. Most Danish pension funds have longstanding expertise in and knowledge of how to invest in Copenhagen but are on shakier ground outside of Copenhagen. It is for this reason that PFA Pension partnered with the regional property developer A. Enggaard on the Stigsborg Havnefront tender. PFA Pension’s main contribution to the project is to finance developments, and it conducts its own housing developments in the redevelopment zone.

Private Investor

Enggaard is the largest developer in Jutland and has undertaken development work in Aalborg for decades, fostering close collaboration with the municipality. A. Enggaard manages developments

in-house and dedicates a significant proportion of its resources in Aalborg to the Stigsborg Havnefront development. Thus, the Stigsborg P/S development corporation is legally a separate organizational entity, but operationally the corporation sits within A. Enggaard. Thus, Stigsborg P/S development corporation's activities are embedded in A. Enggaard's existing structures and institutions, which reduces the cost of operations and ensures that the development corporation has access to a wide range of skills ranging from legal to financial to sales.

In addition to providing highly specialized skills on-demand, this setup also helps keep the operating costs of Stigsborg P/S low, amounting to just DKK 2 million annually, and allows the Stigsborg operations to expand and compress in a flexible and non-disruptive manner. Stigsborg P/S's senior project leader is a longstanding municipal employee who worked out of A. Enggaard from 2018 to 2020, allowing the municipality to gain knowledge about how to conduct developments of this scale and to help strengthen the development partnership, before returning to work for the municipality. As Hessellund says, "The city can finance its own developments, but through the joint venture, it accesses knowledge of how to do so" (interview, 26 June 2020). A. Enggaard's main tasks are to prepare the land for developments and then to sell the land to private investors and developers.

In 2017–2018, the mortgage provider Nykredit worked on a financing plan targeting Stigsborg Havnefront investors and developers. The Nykredit analysis addressed such questions as: How much does it cost to build? What are the loan options? What is the building maturity rate?

Stigsborg P/S's board of directors consists of the chair Flemming Borreskov, who is a former CEO of Realdania and was agreed upon by all three owners; three politicians, including the mayor, selected by the municipality; one individual selected by PFA Pensions; and two board members selected by A. Enggaard.

Governance Restructuring

"We had to ask ourselves: Are the interests of the city properly represented here?" Peder Baltzer Nielsen says, "and we decided to change the governance model a bit" (interview, 11 August 2020).

The head of the secretariat (from 2018 till present day) at the Stigsborg P/S development corporation, Annette Rosenbæk, had been on a leave of absence from her role at the municipality, but it was decided that she should return to her role in the municipality while continuing her work as the head of the secretariat. A new municipal working group, consisting of Annette Rosenbæk, Anton Hessellund, and Charlotte Krogh, was created and tasked with preparing documentation for the corporate management and board. The group was also tasked with coordinating the different actors to cooperate on the local plans, building rights, definitions of quality, materials, etc.

The new working group has formalized procedures to better meet the needs of the municipality. Thus, while a previous private investor that had a close working relationship with Enggaard would comment on and seek to influence the area plan, the new procedure prevents this type of intervention. "The slight restructuring represented a recognition by all three partners that we had a problem that needed fixing," says Peder Baltzer Nielsen (interview, 11 August 2020).

Three members of the city council, including the mayor, sit on the development corporation's board. However, Peder Baltzer Nielsen observed that it would be advantageous if his own immediate supervisor, the politically appointed Councilor, were on the board, too, as this would enable the board to have a better-informed dialogue on the specifics of the Stigsborg development. A significant challenge for the joint venture is simply that the municipality is accustomed to having sole responsibility for constructing public infrastructure, including roads, green areas, and water pipes, but now must engage in joint decision making.

In the joint venture, revenues for public infrastructure come from the yield of the sales and developments, and the development corporation is responsible for the public infrastructure. "At the same time, the development corporation is very willing to push public infrastructure financing and delivery to the municipality," says Peder Baltzer Nielsen. "This will reduce the costs for the development corporation and thus increase the partners' revenue" (interview, 11 August 2020). One example is a promenade estimated to cost DKK 15,000,000 that runs alongside Stigsparken and is financed by land sale revenues gained by the municipality. The two non-municipal joint venture partners argued that the promenade was part of the overall development of the park. "The whole point of the development corporation is that the taxpayers should not pay for Stigsborg's public infrastructure," Peder Baltzer Nielsen says. "If the municipality pays for the infrastructure, then we should maintain our right to plan at a detailed level. At least that is how the municipal traffic planning department sees it" (interview, 11 August 2020).

Ownership

Stigsborg P/S development corporation is a joint venture, of which Aalborg Municipality owns 49 percent and private investors, PFA Pension, and the property developer A. Enggaard jointly own 51 percent (ownership is split evenly between the three parties). The municipality tendered this joint venture so that various parties competed for the right to coown the development corporation with the municipality. PFA Pension and A. Enggaard won the tender in 2018.

According to City Architect Peder Baltzer Nielsen, Aalborg Municipality considered maturing the land and selling it directly to developers rather than establishing the joint venture with PFA Property A/S and A. Enggaard. However, the joint venture proved attractive because: 1) the city wished to have a greater say in how the new Stigsborg development area evolved, and 2) the city wished to capture part of the revenue generated by developing the entire area, including buildings. Nevertheless, Peder Baltzer Nielsen says the city recognizes that it cannot be certain that it will receive more revenues by moving further down the value chain: "Being able to create a unique part of the city and better cohesion across Aalborg as a whole played a decisive role in our final choice of how to proceed" (interview, 11 August 2020). As a partner in the joint venture, the municipality retains greater discretion over the corporation and its developments.

Anton Hessellund says, "We don't get any advantages [read; in terms of being granted favorable planning and building permissions, for instance] from the city being a partner. But of course, the city knows the area and development better than the three other areas (along the four axes of developments)" (interview, 26 June 2020).

Investment

In 2017, Aalborg Municipality contributed to the joint venture with land assets worth DKK 187,500,000. The partners considered the value of land too elevated vis-à-vis the 49 percent that the municipality owns of the joint venture. To compensate, the two other partners, PFA Pensions and A. Enggaard A/S, paid the municipality DKK 68,185,000. The municipality subsequently invested DKK 26,000,000 in a public park called Stigsparken. The costs of the park were considerable in part due to the contaminated soil stemming from the former Kemira plant.

In total, Aalborg Municipality paid DKK 119,315,000 for 49 percent of the shares of the joint venture, while the private partners paid DKK 124,185,000 for 51 percent of the shares. Thus, in total, the joint venture was started with capital of DKK 243,500,000 in 2018.

Prior to maturing the land, the municipality invested DKK 30,000,000 over a two-year period to plan the area development, which included evaluating soil and groundwater quality, assessing financial costs and revenues, and carrying out the tendering for joint venture partners. Nykredit Market managed this process, which also included Vandkunsten (which won the planning competition) and COWI (which conducted the soil and water tests).

Nykredit Market assessed how the municipality could convert the land ownership into building rights and at what value. Peder Baltzer Nielsen says, “We would ask Nykredit Market: How about if we build in bricks rather than concrete? How much does that increase the costs and sales prices? [...] Nykredit gave the green light for raising the building standards” (interview, 11 August 2020).

Peder Baltzer Nielsen says that the joint venture represented a steep learning curve for all three partners. At the start of the joint venture, none of the three partners were quite sure about the municipality’s role. For instance, potential clients, such as private investors and developers, questioned whether they could benefit from doing business directly with the municipality. Initially, even the municipality was uncertain how to react to pressure to accommodate wishes from private investors and developers that sought to water down the original plans for the area. Such conversations occurred in a business setting that represented unfamiliar ground for the municipality.

The municipality had to ask its partners how much it should accommodate potential customers. Peder Baltzer Nielsen believes it is easier to be on the outside regulating how development should occur and feels that the municipality’s role in area development is clear: it defines the area planning, grants permissions, and provides the public infrastructure. While the two other joint venture partners naturally viewed the area development from a business perspective, the municipality needed to stand firm on serving the public good in addition to earning revenue for the city. In short, the joint venture raised questions concerning the role of the municipality vis-à-vis private investors and developers, and it revealed the joint venture partners’ diverse motivations in pursuing the development.

Access to Capital

With the establishment of the Stigsborg P/S development corporation, Aalborg Municipality was able to retain within the corporation the finances associated with the development. The corporation

thus offers annual revenue sharing to the three partners, and its finances are not included in annual municipal budgets. “We’re able to separate the development costs from the annual budgets and decisions concerning how many kindergartens we should build,” says Peder Baltzer Nielsen (interview, 11 August 2020).

Capital for distinct building constructions stems from private investors and developers (including PFA Pension and A. Enggaard). Approved in 2020, the local development plan includes a pedestrian and biking bridge between Stigsborg Havnefront and the city center, estimated to cost DKK 400 million. DKK 100 million will be provided by the municipality through public spending, DKK 100 million will be financed through increased land prices, and the remaining DKK 200 million will come from other sources. COWI undertook an analysis of the costs of maturing the land for development.

Conclusion

Stigsborg Havnefront is an interesting example of how a part-publicly-owned and privately driven development corporation is composed. Generally speaking, the municipality contributed the land assets to the joint venture of the Stigsborg P/S development corporation, PFA Pension (fund) contributes investment capital and is a long-term patient partner, and A. Enggaard is a local private developer contributing granular knowledge of how to conduct large-scale developments in the area.

Rather than the municipality simply handing over the land, the municipality insisted on selling the land and subsequently reinvesting part of the revenue into the partnership of Stigsborg P/S development corporation. In effect, this meant that the municipality became a capital partner on equal terms with the two other partners in the partnership. As a capital owner, the municipality has a direct and equal say on how things are done.

Aarhus

Introduction

Aarhus is the second largest city in Denmark. Greater Aarhus currently has a population of around 350,000, but the city is growing continuously, having increased by 50,000 inhabitants since 2008. The city has a thriving university life with almost 50,000 students. The city employs 20,000 people in its local administration.

This case study concerns the urban redevelopment of Aarhus’ deindustrialized harbor areas. Aarhus is home to another large-scale container port in Denmark, and urban expansion occurs in parallel with the expansion of industrial activity in the container port. This is possible because the container port has moved from the northern side of the harbor, where the city is expanding, to the eastern side of the harbor, where room for expansion is being created through land reclamation, as illustrated in image 37.

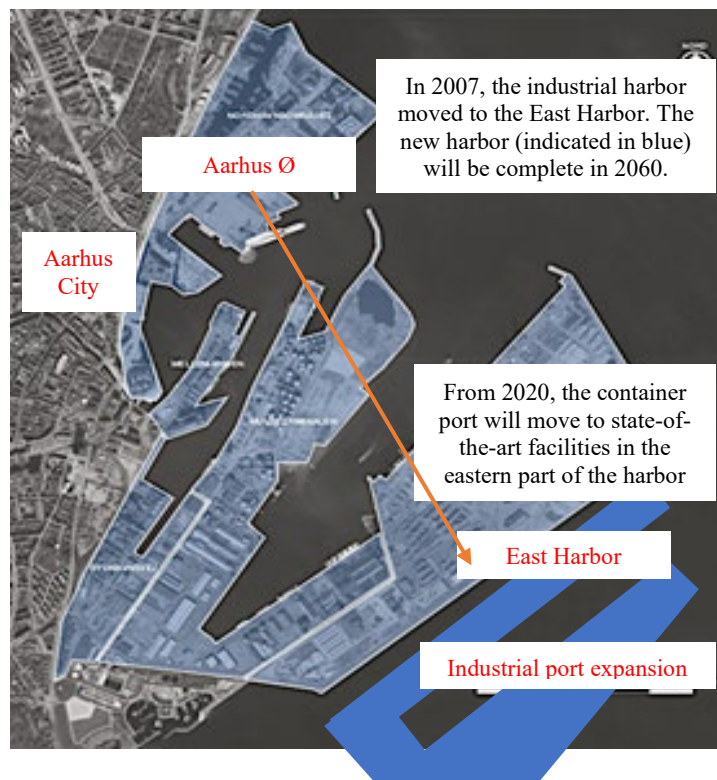


Image 37: Overview of the Århus harbor development area and the move of the industrial harbor. Source: Noring, 2021.

The container port moved from the North Harbor (today's Aarhus Ø) in 2007, and there is no industrial activity in Aarhus Ø today. The industrial harbor is expanding in the East Harbor. The ferry between Jutland and Zealand has also moved from Aarhus Ø to the East Harbor. The South Harbor, which is not represented in the above image, is the latest area to undergo urban transformation.

This case study of Aarhus is largely based on an interview with the Head of City Strategic Developments, Bente Lykke Sørensen (19 February 2020). Lykke Sørensen is responsible for strategic and neighborhood redevelopment, with a focus on the former industrial port areas. The urban redevelopment zone of Aarhus Ø, centered on a former port area (i.e., the Inner Harbor and the South Harbor combined), represents a substantial area of 800,000 floor m². The entire urban redevelopment of the former port areas is scheduled for completion in 2030.¹⁹

Sequencing

Aarhus uses the urban development planning tool developed by Realdania, which was first deployed in 2013 with the redevelopment of Aarhus Ø, the first area to undergo transformation. The unit of strategic urban development, headed by Bente Lykke Sørensen, works within the city administration and closely with the municipal council, which is instrumental in defining the visions for the different areas. In 2013, the municipal council also made recommendations for improving Realdania's urban development planning tool. These recommendations primarily concerned

¹⁹ https://www.aarhushavn.dk/download/nyheder/masterplan_aarhus_havn_2018-06-22_-_medium.pdf.

improving city life. The global financial crisis prompted the city to deploy the Realdania planning tool as a means of ensuring that a vision and strategy are in place prior to embarking upon the material aspects of development. The vision-driven approach deployed in South Harbor is emblematic of this approach and has been enabled by the municipal ownership of the harbor. Because there were existing profitable activities on site (i.e., a power station and a slaughterhouse), there was less fiscal pressure for making a return on investment than was the case for Århus Ø, where the municipality purchased the land from the harbor authority and where the global financial crisis threw plans into disarray.

Aarhus Ø

The urban redevelopment of Aarhus Ø began with the publication of a master plan in 1997. The master plan targeted the North Harbor, which was later renamed Aarhus Docklands, and then Aarhus Ø. In 1997, the harbor authority gave the municipality 600,000 m² of Aarhus Ø land for urban redevelopment. The municipality compensated the harbor authority with new land and a new container terminal in the eastern part of the harbor. The city has thus simultaneously sought to expand urban functions within the deindustrialised port areas and expand industrial port activities in the East Harbor.

The city began redeveloping Aarhus Ø in 2007, which coincided with the onset of the financial crisis. Bente Lykke Sørensen says: “By that time the local development plan was approved, and the developers had received the building permits, the developers were bankrupt!” (interview, 19 February 2020). Lykke Sørensen explains that her team had to use bank guarantees to get half the money the city had been promised. However, as the Aarhus Ø land had already been prepared for construction, the city decided to increase the number of temporary activities, while Bente Lykke Sørensen and her team continued the sales work.

The temporary activities held in the redevelopment area were indicative of what the city hoped to achieve. For instance, Aarhus Ø has a beach café that BIG Architects transformed into a harbor swimming area. In South Harbor, some of the temporary activities morphed into art installations. The city also uses the construction zone’s fencing to tell the story of what the area will look like in the future because, as Lykke Sørensen says, “the value of these area is totally dependent on people’s ability to envisage the future vision and potential of the place” (interview, 19 February 2020).

In 2013, with the construction of the iconic and award-winning 210-unit residential building The Iceberg, new investors started engaging with the city on post-recession developments in Aarhus Ø. However, the new wave of developments was complicated by the fact that former (and often bankrupt) developers had been granted building permits. Bente Lykke Sørensen needed to establish a dialogue between the new and the old developers in order to complete the planned projects.



Image 38: The iconic Iceberg building in Aarhus Ø. Source: <https://hamiconsult.dk/portfolio-items/isbjerget-aarhus-havn/>.

Bente Lykke Sørensen recalls highly negative press coverage during this period, with criticism levelled at the failure to achieve the projected developments within the estimated timeframe, and the city being seen, with the construction of The Iceberg, as being out of touch with the reality of the then-current market situation. Peter Noor Malmø suggests that political leadership, total dedication to the project on the part of municipal politicians and administrators, and the decision to give Bente Lykke Sørensen the necessary leeway to pursue her strategy were all vital to Aarhus Ø's eventual success.

By 2010, developments in Aarhus Ø were back on track, and are now scheduled for completion in 2025–2035, when 12,000 to 14,000 inhabitants will be living in Aarhus Ø. The area adjacent to Aarhus Ø will be completed in 2028–2030.

In the 1980s, South Harbor was already a site for temporary activities. This idea to dedicate sites to temporary activities prior to embarking upon development was followed in the South Harbor. In the Århus Island, this model of offering temporary activities was copied between 2015 and 2017.

South Harbor

Early developments in the South Harbor commenced in 2017 with plans for the preservation and renovation of old historic buildings and the creation of new buildings. The South Harbor only contains 7,000 workspaces, but the neighborhood marks a shift in the way in which the city is conducting urban development. Only commercial and office buildings will be permitted in the development zone as it is too close to the industrial harbor—and thus prone to noise, smells, and air pollution—for residential use. Unlike Aarhus Ø, which had no existing residents for whom the city needed to take responsibility, the South Harbor has a history of commercial and industrial use, including a power station and a slaughterhouse, in addition to an existing non-residential population, including many socially vulnerable individuals, artists, and musicians who used vacant buildings close to the city proper for their activities. The city thus integrated processes of resident engagement, vision cocreation, and community strategy work into the South Harbor redevelopment

project, which will be complete in 2030–2035. In the words of Lykke Sørensen, “When we start the construction project after four years [of resident engagement], we know exactly what we are doing. We work with social diversity and take social responsibility for the existing population living in the areas” (interview, 19 February 2020). In South Harbor, for instance, there are many homeless people, to whom the city delivers food. During this process, every landmark building presents a new opportunity for resident engagement. “Over the years,” says Lykke Sørensen, “we’ve become very good at establishing a clear vision and being specific about what we want from private investors and developers” (interview, 19 February 2020).

South Harbor is located nearer the city center and adjacent to the industrial harbor. The central station can be reached in five minutes by foot. South Harbor will eventually cover 120,000 floor m² but is still undergoing master planning, which should be completed in 2022.

The timeline of image 39 illustrates the sequence of events of developments in Aarhus port:

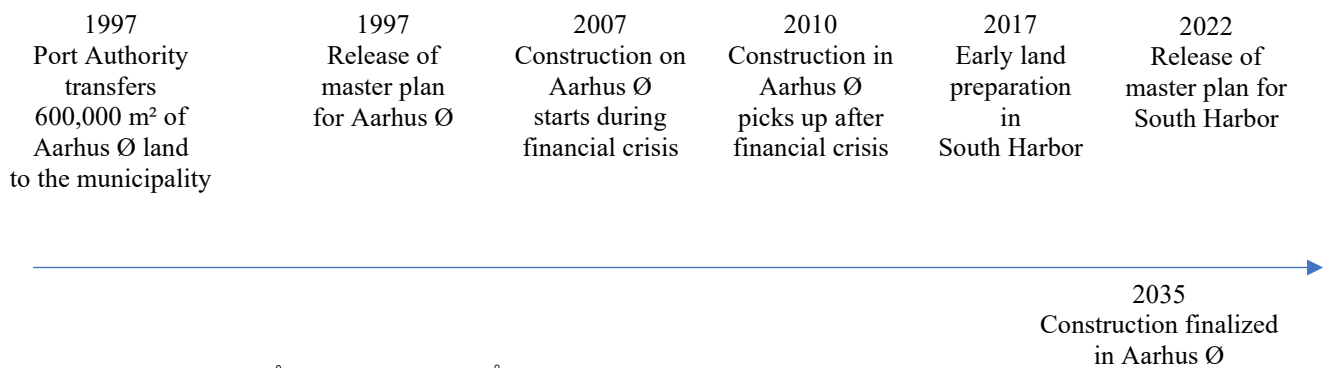


Image 39: Timeline of Århus Island (Read *Århus Ø*) development. Source: Noring, 2021.

Strong Vision

As discussed above, the city has assigned different characteristics to each of the areas under development. In Aarhus Ø, with no existing residents, the city could freely run competitions for the master plan, local development plans, and iconic buildings, whereas the South Harbor possessed a range of existing social, artistic, and cultural activities. These characteristics are incorporated into the new urban vision for the area. The vision for the South Harbor is to create a dense and diverse neighborhood with space for artistic and cultural activity and expression.²⁰ There will also be space for start-ups.

Aarhus Ø redevelopments include 25 percent affordable and social housing, which is in line with the city’s strategy of ensuring a steady supply of affordable housing. In addition, the city is eager to provide affordable housing to its growing student population. The desire to include affordable and social housing reflects the city’s political vision (created by the Municipal Council) and urban redevelopment strategy (shaped and implemented by Lykke Sørensen and her team) as well as the fact that residential housing was crucial for the city’s recovery from the financial crisis.

²⁰ <https://sydhavnskvarteret.dk/media/7295/udviklingsplan-for-sydhavnskvarteret.pdf>

Aarhus is using its land ownership to encourage the development of affordable housing in the city, a process that involves negotiations between the municipality, private developers, and private nonprofit housing associations, occurring within a national regulatory framework.

Lykke Sørensen explains, “We collaborate with the municipal council and the market actors. If the market alone decided, the municipal council and residents would not get the city we want” (interview, 19 February 2020). The municipal council, which holds elections every four years, represents the community councils and all the residents of Aarhus, and it gathers input through public hearings and other events, thereby imbuing the planning process with democratic legitimacy.

“The challenge is that the municipal council needs to handle both the role of political visionary and the role of project developer that generates economic value. They have to navigate between politics and markets. It is a powerful combination, but difficult to manage,” concludes Lykke Sørensen (interview, 19 February 2020).

Organizational Structure

Aarhus Municipality’s Department of City Strategic Developments, which is headed by Bente Lykke Sørensen, is the city’s in-house development corporation, with 35 employees who work on municipal planning, including thematic and area plans. Lykke Sørensen and her team are integrated into the municipality’s wider Department of Technology and Environment. This department’s number of staff was reduced from around 1,300 to 1,000 employees in 2020 when the municipality privatized a formerly municipal-owned renewable energy company.

The Department of Technology and Environment is closely tied to the Mayor’s office. Thus, the mayor, the city director, the municipal finance director, and the director of technology and environment all oversee Lykke Sørensen and her team.

Ownership

Aarhus has not yet entered into any joint ventures, as the city is the sole owner of the land assets under development. Sometimes the city accumulates these assets by purchasing land from other public and private owners, before preparing it for sale to private investors and developers. Such municipal land purchases have made possible the development of 3 million floor m² of commercial space. The city is hereby accumulating land assets for future developments. In 2015, the department also paid DKK 400 million for a disused hospital, which had been sold off as a part of a regional agglomeration process and is located in the Tage-Hansens Gade development area.

Investment

As described above, the neighborhoods arising out of urban redevelopment are a result of the municipal council’s visions and policies and the capability of Bente Lykke Sørensen and her team to put these into action. Lykke Sørensen explains: “The politicians want densification of the city center. But we have to take into account market conditions. What do the residents want? We need to consider all three stakeholder groups: politicians, market actors, and residents” (interview, 19 February 2020). In addition to resident workshops and public hearings, Lykke Sørensen and her

team facilitate market dialogue in the form of general consultations with market actors. Such consultations do not concern specific projects. Market dialogues concern themes such as communal living for the elderly and other hot political topics. These dialogues take the form of networking events, workshops, city tours, visits to food halls, etc., in which all investors and developers are invited to participate.

In addition to the five large areas that are being developed by the Department of City Strategic Developments, the department also manages a series of smaller development zones.

The Department of City Strategic Developments manages the five large redevelopment areas, including the hospital (101,000 floor m²), Aarhus Ø (800,000 floor m²), South Harbor (120,000 floor m²), Lisbjerg (1,000,000 floor m²), and the freight yard area (130,000 floor m²).

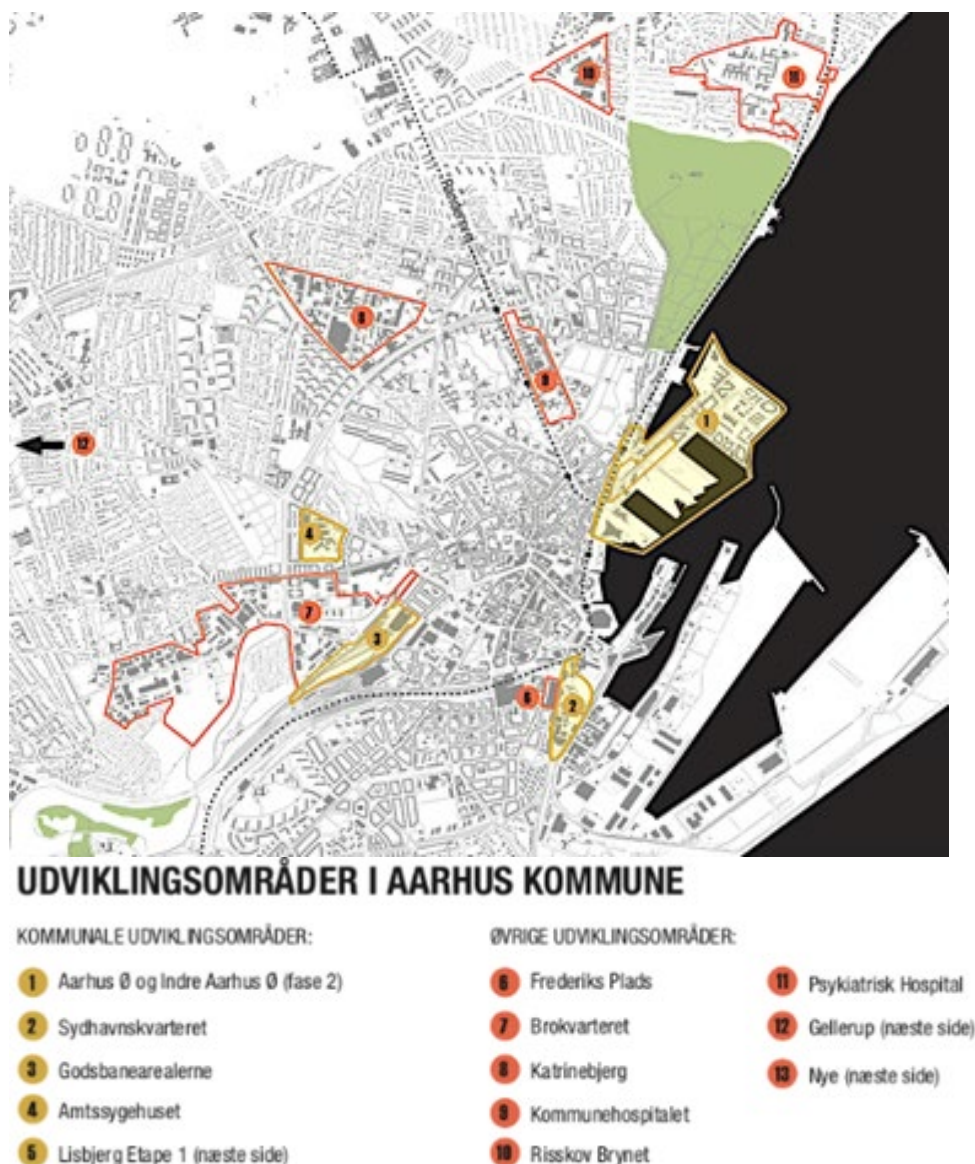


Image 40: Overview of development areas of Århus. (*Areas 5 and 13 are not presented on the above map, as they are outside of the city center). Source: Department of City Strategic Developments, 2020.

The Tendering Process

The tendering process creates the transparency demanded by private market actors. Any private investor or developer can join the tendering process, and the final decision relies upon a professional—rather than a political—evaluation.

Developments in Basin 7 of Aarhus Ø have been emblematic of how the municipality conducts tendering. When the municipality conducts a tender, it selects five final proposals and developers. The city then engages in dialogue with all five developers on how they can optimize their proposals and work toward one “super proposal.” Those developers whose ideas are integrated into the final super proposal are well compensated. During this process, the developers work with the City Architect and are offered time to modify their proposals. A professional evaluation panel of judges, the magistrate, and the municipal council reach the final decision on the winning proposal.

In some cases, the city may proceed to the intermediary step of granting a conditional right of purchase to the developer. In this approach, the city conditionally sells the land before the local development plan has been determined. As Lykke Sørensen explains, “We use the local development plan as a management tool, where we co-develop the area with the private developer” (interview, 19 February 2020). Conditional rights of purchase mean that the city and developer work together over the course of nine months to refine the local plans and building developments. This grants sufficient time for negotiating the right agreement with the developer, including going back to the municipal council with questions and clarifications. Lykke Sørensen and her team thus have the opportunity to shape projects that are perceived as crucial to the overall neighborhood redevelopment. This is a time-consuming endeavor, but it ensures that projects align with the city’s vision. Thus, price is not the most important factor in these decisions, as the city prioritizes realization of its urban development vision. In the phase of building and realizing the project, Lykke Sørensen and her team coordinate with the developer on activities that relate to preparing the land for development.

Image 41 shows the South Harbor area as it looks today:



Image 41: This image shows the South Harbor area as it looks today. Source: Luise Noring, February 2020.



Image 42: The South Harbor redevelopment project involves the city partly repurposing the train tracks, as pictured in the urban planner's rendering. Source: <https://sydhavnskvarteret.dk/media/7295/udviklingsplan-for-sydhavnskvarteret.pdf>.

Access to Finance

Aarhus Ø represents an accumulated investment of DKK 3.8 billion from the municipality (excluding land purchase costs) and around DKK 20 billion from private investors.

All Danish municipalities operate with a fixed loan ceiling (Read *anlægsloft*) that prevents them from borrowing in excess of DKK 230 million annually and encourages prudent local public spending. During the recent COVID-19 crisis, the national government increased the fixed loan ceiling to DKK 500 million annually in order to incentivize local public spending. Nevertheless, due to the fixed loan ceiling, many of the development projects described here could never have been realized if the city had been unable to raise investment capital through sale of its land assets. This capital is subsequently reinvested into public land assets to prepare more land for private developments. This method is termed 'land value capture.' It emphasizes the importance of ensuring that private developments on public land reflect the city's vision. Thus, a development process that strongly aligns with the city's vision has the potential to not only drive urban expansion but also reinforce the city's core values. The strong vision and values allow the city to charge a price in excess of the value of the land. In this way, the city exchanges immediate financial gain in individual projects to achieve this vision, which works as a multiplier, increasing the quality of the development areas over the medium term and long term, and bringing its own financial benefits. Close collaboration between the municipal council, the Strategy Development Team, and private investors and developers through an efficient planning and development process enables new urban areas and innovative solutions.

The potential of land value capture and its underlying financial mechanisms to a large extent rely upon the government's desire and ability to reinvest the capital raised through land sales in a revolving funds model. That is, the government must refrain from extracting profits and diverting

funds. Land value capture is particularly suitable in cases in which the government does not wish to use tax-funded capital to invest in urban development yet acknowledges the need for new developments in order to continuously grow and transform.

Using land value capture to drive urban expansion in line with the city's vision requires that Lykke Sørensen and her team strike a balance between market demands and the political demands of the municipal council. "If the investors sense that there's too much politics, they run away screaming. It's essential to our success that the municipal council is capable of keeping the political discussions in the political realm and let the developers get on with their jobs," Lykke Sørensen concludes (interview, 19 February 2020).



Image 43: Detailed overview over the buildings of Århus Island. Source: www.aarhusoe.dk.

Conclusion

Aarhus represents a complex case due to the different points of departure and visions for the five development areas. The city and its Strategy Development Team, headed by Bente Lykke Sørensen, have demonstrated the ability to adjust and adapt in response to these differences.

The redevelopment of Aarhus' harbor commenced in a challenging economic climate but has succeeded in bringing together the necessary capital, actors, and—significantly—vision for creating diverse and vibrant city center neighborhoods. These achievements have been supported by

dedication to a revolving funds model of financing, relying on land value capture and tendering with conditional rights of purchase to support affordable and high-quality development.

Conclusion

A Strong Vision

It is crucial to have a vision of quality, rather than following strict market logics of maximizing sales prices of individual parcels. Danish towns and cities realize that land is of limited value without vision and purpose. Development corporations focus the private sector towards driving innovations that increase land value through multistage competition processes, and this value is then converted into projects that continue to fulfil the goal of creating visionary towns and cities.

The development corporations exchange immediate financial gain for the ability to create developments that fulfil a town's vision, which increases the overall urban quality of the development areas over the medium and long term, producing its own financial benefits. Future residents and companies purchase and choose to live in and engage with spaces that resonate with their lifestyles, ideals, and ambitions; as a result, a significant share of the value of an apartment or office building comes from the quality of its immediate surroundings and the character of the town or city as a whole. Land is not simply a plot, a parcel, or a commodity; it is transformed into a crucial tool for achieving urban quality and providing public goods. Articulating a vision of the town or city and then embedding this vision in the urban fabric through continued public reinvestments and collaboration with private stakeholders increases land value over the course of development.

The development corporations build upon land value capture strategies by using their pivotal positions at the nexus of institutions, the urban planning process, and finance to pursue strong and unique visions of urban quality. In contrast to strategies that regard income generation as an end in itself, the public infrastructure investments, standards setting, and competitions managed by the development corporations seek to convert unrealized income into the urban quality encompassed by the vision. As projects are completed, towns and cities forgo the opportunity to pocket potential profits, instead allowing the development corporations to reinvest profits to reinforce the vision that will allow them to generate even greater profits from future projects. Emphasis is thus placed on long-term value creation for the town or city.

Through the accumulated creation of urban quality across multiple development projects, towns and cities are implementing their visions and shifting to uncontested niche market positions to the mutual benefit of private investors and developers as well as companies and residents in the redeveloped areas.

Land Value Capture

Another important commonality is that all these cases involve the bundling of publicly owned land and building assets under the management of one organizational unit that is granted considerable freedom of operation in the market. This operational freedom both ensures that the development

corporations are not derailed by shifting political priorities and grants each organization the agility required to operate in the market, including optimization of asset prices. They all leverage the market to generate financing for urban redevelopment; the development corporations use land value capture to boost the value of public assets through rezoning and infrastructure investments. Land value capture is a finance mechanism available to public organizations that own and/or control public land, possess rezoning authority, and have the financial resources to invest in large-scale infrastructure projects. The development strategy of land value capture relies upon investment to increase land values.

The potential value of these areas arises from their location. Many areas developed through this strategy are adjacent to existing urban areas or have been incorporated into the city, as urban expansion has subsumed the former periphery. In the standard path for land value capture, former industrial spaces are rezoned for residential and commercial purposes, which increases their value. Governments subsequently sell the land at the increased value and use the yields to finance infrastructure, which further raises the value of the land under development. In order to finance infrastructure upfront, loans can also be taken out against the projected increased value of the assets. Any additional yield from the land sales is used to service debt, make further investments, or boost government coffers.

Replication in Other Cities

Public ownership of land is beneficial. However, even cities that do not directly own land in redevelopment locations can still deploy many of the strategies described in this working paper. Municipalities retain the ability and authority to create masterplans, draw zoning boundaries, and award building permissions. By raising urban quality through a defined vision, cities can direct private investments and developers toward urban growth that serves greater development goals, including environmental and social benefits. The incremental creation of value increases the city's negotiating position in its efforts to direct redevelopment to fit its vision. By using the public sector to formulate an urban vision and then maximizing the potential of collaboration with private investors and developers to support and develop this vision, cities around the world can realize socially, environmentally, and financially sustainable urban transformation.

The key facilitating condition is the recognition of the value of urban vision for translating goals into built, material outcomes. Not all visions need to be the same as the ones presented in this working paper; towns and cities should shape their visions and goals based on their existing values and characteristics.

List of Interviewees

In addition to desktop research, this working paper is largely the result of extensive interviews and continuous dialogues with the following people:

Name	Title	Company and municipality	Dates in 2020
Claus Ravn	Lead Consultant	Realdania By & Byg	5 March
Ole Møller	CEO/ Project Director	Nærheden P/S, Høje-Taastrup	27 January
Bente Lykke Sørensen	City Strategic Director	Aarhus Municipality	19 February
Michael Tolstrup	Head of Department	Aarhus Municipality	3 December
Mia Hørup	Special Consultant	Aarhus Municipality	3 December
Claus Omann Jensen	CEO / Director	Færchfonden, Holsterbro	19 February
Hanne Torp Brodersen	Project Director	Færchfonden, Holsterbro	19 February
Peter Frost	Municipal Director	Køge Municipality	30 June
Tove Skrumsager Frederiksen	Project Director of Køge Kyst P/S	Køge Municipality	10 December
Hans Østergaard	Mayor	Ringkøbing-Skjern Municipality	24 June
Anton Hessellund	Director	Stigsborg Havnefront P/S, Aalborg Municipality	26 June
Tim E. Halvorsen	CEO / Director	Naturbydelen in Ringkøbing + Kanalbyen in Fredericia	10 June
Peder Baltzer Nielsen	City Architect	Aalborg Municipality	11 August